Do Firm Credit Constraints Impair Climate Policy?*

MATTHIAS KALDORF[†] MENGJIE SHI[‡]

March 11, 2026

Abstract

Credit-constrained firms respond less to climate policy shocks. Empirically, firms with lower distance-to-default exhibit a smaller emission reduction after a carbon tax increase. We incorporate this empirical finding into a quantitative DSGE model with tractable firm heterogeneity, credit frictions, climate policy, and emission abatement investments. Consistent with the data, the model gives rise to a novel credit demand side channel that is associated with a debt overhang problem and is not driven by credit supply. Quantitatively, achieving net-zero emissions requires carbon taxes that are approximately 2\$/tCO2 higher for constrained firms. Removing the negative effect of borrower-driven credit frictions on abatement investment, for example, through a green credit guarantee scheme, has a positive but small effect on welfare.

Keywords: Climate Policy, Credit Constraints, Emission Reduction, Corporate Capital Structure, Firm Heterogeneity

JEL Classification: E22, G31, G32, Q54

---

*Aymeric Bellon, Tobias Berg, Ricardo Correa, Claudia Custodio, Dominik Damast, Francesca Diluiso, Min Fang, Maren Froemel, Emilia Garcia Appendini, Martin Goetz, Galina Hale, Karol Kempa, Emanuel Moench, Alexander Popov, Peter Raupach, Matthias Rottner, Christian Schlag, Bram van der Kroft, our discussants Volha Audzei, Jon Frost and Nora Wegner and participants of seminars at Deutsche Bundesbank, Goethe University Frankfurt, the 2024 Sustainable Finance PhD workshop (Augsburg), Chinese Economic Association UK (London), Econometric Society European Summer Meeting (Rotterdam), 2025 American Economic Association Meeting (San Francisco), 2023-2025 IBRN Annual Meetings (Basel, Mexico, Oslo), 2025 ESCB Research Cluster Climate Change, 2026 American Finance Association Meeting (Philadelphia) provided useful comments and suggestions. The views expressed here are our own and do not necessarily reflect those of the Deutsche Bundesbank or the Eurosystem.

[†]Deutsche Bundesbank, Research Centre. Wilhelm-Epstein-Str. 14, 60431 Frankfurt am Main, Germany. Email: matthias.kaldorf@bundesbank.de (Corresponding Author).

[‡]Deutsche Bundesbank, Research Centre and Goethe University Frankfurt. Email: mengjie.shi@bundesbank.de.

1 Introduction

The net zero transition requires large investments in low-emission technologies. Since firm-level investment in green technologies relies heavily on debt financing, this immediately raises the question of whether frictions associated with firms' debt financing impair the transmission of carbon taxes to emissions at the firm level. Answering this question is crucial for the design of climate policies, as policymakers need to take the role of firms' financial frictions into account along the transition to net zero.

Empirically, we document the relevance of firm credit constraints for climate policy in a large international sample of listed firms from 2012 to 2019. We combine firm-level emissions data with their distance-to-default (Merton, 1974), which has been identified as a suitable measure of credit constraints by Farre-Mensa and Ljungqvist (2015). To measure climate policy, we employ an annual dataset of country-specific carbon taxes, which is maintained by *OECD Statistics* for 33 countries that jointly accounted for 77% of global emissions in 2019. Employing a standard difference-in-difference approach, we find that firms with tighter credit constraints, i.e., a lower distance-to-default, cut their emissions less than their unconstrained peers in the same industry in response to a carbon tax increase. Quantitatively, a one–standard-deviation increase in distance-to-default is associated with a 1.1 percentage point reduction in emissions growth in the full sample and a 2.74 percentage point reduction in the manufacturing sector. This effect is economically sizable, given that the standard deviation of annual emissions growth is 0.91 percentage points in the full sample and 0.87 percentage points in the manufacturing sector.

Motivated by this empirical evidence, it is our main contribution to assess the implications of borrower-driven credit constraints for climate policy and welfare in a quantitative E-DSGE model. The model combines elements from the environmental business cycle literature (Heutel, 2012) with financial frictions in the manufacturing sector (Gomes, Jermann, and Schmid, 2016). There are two types of manufacturing firms that are owned by households and use identical production technologies but choose to finance their investment in different ways. Manufacturing firms are subject to uninsurable idiosyncratic productivity shocks. They generate carbon emissions as a by-product of their production and are subject to carbon taxes.

Firms have the option to issue defaultable debt to or raise equity from households. The first type of manufacturing firm is managed by patient managers who have the same time discount factor as households. Such firms have no incentive to issue debt and turn out to be *safe*. Their optimal abatement effort is solely determined by carbon taxes and the cost of emission abatement. Managers of the second (*risky*) type are more impatient than households and issue debt to front-load dividends. However, firms expose themselves to default risk due to the uninsurable productivity shocks. We assume that they default on their debt if production revenues net of carbon taxes fall short of repayment obligations. Corporate default entails a cost that is borne by creditors and fully reflected in borrowing conditions, i.e., the price of corporate debt. Under their optimal capital structure, risky firms issue debt until the marginal default cost equals the benefits of additional debt issuance. These benefits are given by the relative impatience of firm owners vis-a-vis debt owners, which is the key credit friction in our model and gives rise to endogenous credit constraints.

Credit constraints have direct implications for manufacturing firms' decision to abate emissions. While optimal abatement increases in the expected carbon tax, it also takes into account that firm owners do not receive the payoff from abatement in the case of default. All else equal, the benefits of acquiring abatement goods are lower for risky firms, which reduces their optimal emission abatement. This is precisely where firm credit frictions impair the transmission of carbon taxes in our model.[1] Importantly, relative impatience induces firms to issue debt, which results in a higher default probability and makes investment less attractive due to the classical debt overhang problem (Myers, 1977). Our mechanism does not require that firms stick to conventional technologies that generate large short-term payoffs (coal plants) instead of investing in technologies that are more profitable in the longer term (renewable energy).

We calibrate our model to match key empirical regularities of corporate leverage and default risk over the business cycle. Technology and climate policy parameters are set to standard values in the environmental macroeconomics literature. The model replicates important untargeted second moments, including a counter-cyclical default frequency, pro-cyclical debt issuance and leverage, as well as pro-cyclical emissions, consistent with the data. In the model, firms' credit constraints are furthermore associated with a high default rate and hence a low distance-to-default, which provides a direct link to our empirical analysis. We exploit this link to further corroborate the external validity of our calibration. Specifically, we compare the effect of an unanticipated and permanent carbon tax increase from zero to 10\$/tCO2 on safe/unconstrained and risky/constrained firms, mimicking our empirical strategy. It turns out that emissions of unconstrained firms decline by an additional 0.5 percentage points, which is closely aligned with our empirical findings. This shows that our model quantitatively reconciles the interaction between credit frictions and climate policy that we have documented in the data. We also demonstrate that the relative importance of firm credit constraints is independent of parameters in the climate block, but diminishes once financial frictions become less

[1] The crucial difference to related models building on Heutel (2012) lies in our assumption that investment in abatement goods is chosen before idiosyncratic productivity shocks realize. In models where the abatement effort is a contemporaneous choice, it is not directly affected by credit constraints, see for example Carattini, Melkadze, and Heutel (2023).

severe.

The model's ability to replicate the differential response of constrained and unconstrained firms to *unanticipated* tax shocks corroborates the model's external validity. Climate policies, however, are inherently concerned with long run issues. Hence, we turn to the implications of credit constraints for the firm-level abatement effort in the long run. Specifically, we assess how much firm credit frictions raise carbon taxes consistent with net zero emissions. In our baseline calibration, the full-abatement tax is 94$/tCO2 for safe/unconstrained firms, while it is slightly larger (96$/tCO2) for risky/constrained firms. While this difference appears to be small, it is worth noting that the coverage-weighted average carbon emission price amounts to merely 5$/tCO2.

So far, we have studied the role of credit constraints for climate policy at the firm level by comparing the responses of risky/constrained firms to safe/unconstrained firms. As a final step, we use our calibrated model to assess the relevance of credit frictions for climate policy at the *aggregate* level, and for the welfare gain of carbon taxes. We find that the welfare gain of increasing the carbon tax from zero to 10$/tCO2 in our baseline economy amounts to 1.6% in consumption equivalents. In a counterfactual economy where constrained firms choose the same abatement effort as safe firms, welfare increases by an additional 0.01%. Importantly, we do not remove credit frictions entirely by making both firms as patient as households. Credit frictions still imply default for the risky/constrained firm type and reduce their investment. Consequently, the welfare difference solely stems from the limited transmission of carbon taxes to aggregate emissions.

Importantly, it is possible to make risky/constrained firms choose the same abatement effort as unconstrained firms by a state-dependent transfer to risky/constrained firms. Such a transfer needs to pay the benefit of abatement investment in the case of default to the owners of risky/constrained firms, for instance, through a "green credit guarantee scheme". This reverses the negative effects of frictions on the credit demand side, which result in default risk and debt overhang, on abatement investment. Quantitatively, the additional welfare gain of such a policy amounts to merely one percent of the overall benefit from increasing carbon taxes by 10$/tCO2.[2]

It is worth noting that our credit demand side mechanism can be distinguished from the predominant focus on credit *supply* frictions in the existing literature. For instance, banks' reluctance to lend to carbon-intensive firms (Kacperczyk and Peydro, 2022), the risk of stranded assets in banks' loan portfolios (Covi et al., 2025), or the "greenium" in corporate borrowing (Berg et al., 2025, Hale et al., 2025). These frictions imply that pol-

[2]We do not model how such a green credit guarantee scheme can be implemented through financial markets in practice. While we consider the simplest possible case of direct transfers to firm owners, banks could also receive the state-dependent transfer ex post, which would affect firm owners ex ante through improved borrowing conditions.

luting firms underinvest in abatement because investors are unwilling to extend credit or require a premium. In our framework, constrained firms optimally underinvest in emission abatement not because credit is more expensive or inaccessible, but because default risk and debt overhang reduce the *expected payoff* from abatement investment. As a result, even in the absence of any credit supply restrictions, firms' capital structure decisions impair the pass-through of carbon taxes to emissions. This mechanism complements, rather than competes with, the supply-side literature. Notably, existing studies generally find modest effects of "green" credit supply on emissions.[3] Consistent with this evidence, our analysis suggests that policies targeting the credit demand side, such as green credit guarantees that insure abatement returns in default states, have effects that are economically significant for policy design, though modest relative to the carbon tax itself.

Related Literature There is a growing body of research studying the interactions between credit constraints and emission reductions at the firm level. Goetz (2019) shows that a shock to the debt financing cost increases firms' abatement activities. Accetturo et al. (2023) demonstrate that firms increase their green investment when experiencing a positive credit supply shock. Xu and Kim (2021) and Bellon and Boualam (2024) show that credit-constrained firms are less likely to engage in pollution abatement, which is consistent with our empirical findings for climate policy. Fang, Hsu, and Tsou (2023) study the role of financial constraints for pollution abatement in the context of regulatory uncertainty. Döttling and Lam (2024) use high-frequency identified monetary policy shocks and find that emission intensities slightly increase over the medium term, which is consistent with the notion that tighter credit constraints - due to restrictive monetary policy - induce firms to forgo investment into abatement technologies. De Haas et al. (2024) show that both credit constraints and weak corporate management indicators are associated with higher firm emissions.

Our paper is also related to the recent literature that studies credit constraints, investment and endogenous climate policy in a joint framework. Analytically tractable results are provided by Döttling and Rola-Janicka (2025), Heider and Inderst (2022) and Haas and Kempa (2023). While our analysis delivers novel analytical results, we also contribute to the quantitative E-DSGE literature by explicitly introducing credit frictions into firms' abatement decision. Our framework is related to Carattini, Melkadze, and Heutel (2023), who discuss the macroeconomic effects of asset stranding, and to Giovanardi and Kaldorf (2025), who study optimal bank capital regulation in a multi-sector E-DSGE model with firm and bank financing frictions. Iovino, Martin, and Sauvagnat

[3]Green credit supply is conceptually different from green investing through equity markets, which has empirically been associated with positive effects on firm-level emissions, see for example De Haas and Popov (2022) or Bams and van der Kroft (2025).

(2023) provide an analysis of asset-based borrowing constraints in a general equilibrium model with heterogeneous firms and corporate taxation. Campiglio, Spiganti, and Wiskich (2024) add financial constraints to a directed technical change model and study the interactions between experience effects and the financing of new low-emission technologies. Using a model of external financing constraints that is tailored to small firms, Schuldt and Lessmann (2023) show that relaxing firm-level credit constraints has positive side effects on climate policies. Our paper contributes to this literature by demonstrating that credit frictions have a negative effect on emission abatement through a debt overhang channel that appears to be particularly suitable for studying large firms, consistent with our empirical approach.

Outline This paper is structured as follows. Our data and the main empirical results are presented in Section 2. In Section 3, we introduce an E-DSGE model with heterogeneous firms and endogenous credit frictions. The model's calibration, fit to the data, and implications for welfare and climate policy are discussed in Section 4. Section 5 concludes.

2 Credit Constraints and Climate Policy

In this section, we present evidence on the relevance of credit constraints for the transmission of climate policy to emissions at the firm level. We first outline how we measure climate policy, carbon emissions, and credit constraints, then discuss our empirical specification and results.

2.1 Data

Measuring Climate Policy A key empirical challenge lies in the measurement of climate policies, which encompasses a large variety of different policies, such as emission trading systems, direct taxation, or mandatory industry standards. To align our empirical analysis as closely as possible to our macroeconomic model, we use a country-specific measure of carbon tax stringency, provided by *OECD Statistics*. Data are available for 27 OECD member countries as well as Brazil, China, India, Indonesia, Russia, and South Africa at an annual frequency.

The key advantage of this dataset is that carbon taxes can be readily compared across countries. The carbon taxation index ranges from 0 (no policy in place) to 6 (most stringent). Here, the index value of 6 is assigned to country-year observations above the 90th percentile of the distribution over all countries from 1990 to 2020.[4] Index values

[4]See Section 2.2 in Kruse et al. (2022) for a detailed description.

between 1 and 5 correspond to 10\$/tCO2 intervals. An increase of the index by one point, thus, corresponds to an increase of approximately 10\$/tCO2.[5] The index passes a battery of plausibility checks and is, for example, negatively correlated with country-wide emission-to-GDP ratios. For details on the scope of the dataset and its construction, we refer to Kruse et al. (2022). Section A.1 shows the country-specific time series of carbon tax shocks (Figure A.1) and the evolution of carbon tax levels (Figure A.2) over our sample period from 2012 to 2019.

Measuring Carbon Emissions We obtain firm-level emissions data from *Institutional Shareholder Services* (ISS). The dataset contains annual information on firm-level greenhouse gas (GHG) emissions, differentiating between scope 1 (direct emissions from operations of affiliates that are owned or controlled by the company) and scope 2 (emissions from the consumption of electricity, heat or steam) emissions. We use scope total emissions in our analysis, defined as the sum of scope 1 and scope 2 emissions. Throughout the analysis, we do not take Scope 3 emissions into account, since these refer to activities outside the direct control of the firm. GHGs are defined according to the GHG Procotol, a collaborative accounting standard by the World Resources Institute and the World Business Council on Sustainable Development. Emissions data are either reported or estimated by ISS. Figure A.3 in Appendix A.2 shows how carbon emissions vary across countries and time. In each country, the median emission growth hovers around zero, but sharply declines in 2016 and 2017 after the Paris Agreement was signed. Table A.1 provides summary statistics across countries and sectors. When pooling all observations over time, the distribution of emission growth over firms is fairly symmetric around zero.

Measuring Credit Constraints and other Firm Characteristics Data on credit constraints and firm financial characteristics are from Compustat North America and Global, which assembles data on securities and firm-level accounting information. To proxy for credit constraints, the empirical corporate finance literature has developed a range of indicators constructed from firm-level accounting data, see Kaplan and Zingales (1997) and Whited and Wu (2006) among others. However, as demonstrated by Farre-Mensa and Ljungqvist (2015), firms classified as constrained by indicators based on accounting ratios do not behave as if they were actually credit constrained. Specifically, Farre-Mensa and Ljungqvist (2015) use business tax reductions, which incentivize corporate debt issuance due to the tax deductability of interest expenses, as an exogenous shock to the optimal capital structure trade-off. They show that firms classified as constrained according to several commonly used indicators do not behave differently from

[5] Note that the index does not capture an increase in the *scope* of carbon policies, for example, by requiring more firms to participate in a cap-and-trade scheme.

firms classified as unconstrained.

Therefore, we measure credit constraints by the *distance-to-default (D2D)* proposed by Merton (1974), which passes the plausibility tests for measures of credit constraints proposed by Farre-Mensa and Ljungqvist (2015). Since computing *D2D* requires equity and balance sheet data, this naturally restricts our sample to listed firms. We compute *D2D* for each firm and year closely following the methodology outlined in Gilchrist and Zakrajšek (2012). Figure A.4 in Appendix A.3 shows the evolution of the country-specific average distance-to-default over time, with a full sample average of around eight and a substantial dispersion across countries. Table A.2 contains summary statistics across countries and sectors. There are no significant differences in *D2D* among sectors. Table A.3 also presents descriptive statistics for control variables, whose mean and standard deviation are comparable to those reported in the corporate finance literature (for example, Almeida, Campello, and Weisbach, 2004).

While matching emissions data obtained from ISS, credit constraints data from Compustat, and the carbon taxation index from OECD is conceptually straightforward, we have to make an assumption on the relevance of country-specific carbon policy for multi-country firms. In cases where there are multiple subsidiaries of a given firm in the firm-level dataset, we use its primary location in Compustat, i.e., the country of its headquarters, as the matching entity.[6] Lastly, we exclude financial firms (SIC-codes between 60 and 67, or 69) and public administration (SIC-codes between 90 to 99), as their investment and emissions decisions are not governed by the same debt-financing and credit-constraint mechanisms that underpin our analysis of carbon taxes and firm-level emissions. The final sample consists of 18,882 firms from 2012 to 2019 and 97,913 firm × year observations.

2.2 Empirical Strategy

In our main specification, we examine whether firms' credit constraints affect the pass-through of carbon taxes to firm-level emission growth. Specifically, we regress emission growth on firm j's lagged distance-to-default, changes in the country-specific carbon tax, and their interaction:

$$\begin{aligned}\Delta \log(Emi)_{j,t} = \beta_0 + &\beta_1 \cdot \text{D2D}_{j,t-1} \times \Delta\text{Tax}_{c(j),t} + \beta_2 \cdot \text{D2D}_{j,t-1} + \beta_3 \cdot \Delta\text{Tax}_{c(j),t} \\ &+ \beta_4 \cdot X_{j,t-1} + \chi_c + \delta_i \times \tau_t + \epsilon_{j,t} \,. \qquad (1)\end{aligned}$$

[6]Our empirical findings also hold in the Transportation and Public Utilities sector, as reported in Table B.4, where firms primarily operate domestically and are therefore directly exposed to domestic climate policies.

The dependent variable in equation (1) is firm-level emission growth, defined as the log change in firms' total emissions (in tCO2) from year $t-1$ to t ($\Delta log(Emi)_{j,t} \equiv \log(Emi_{j,t}) - \log(Emi_{j,t-1})$). We use the lagged distance-to-default ($D2D_{j,t-1}$) as a measure of firm credit constraints, since the current distance-to-default might be affected by the change of climate policy and can, thus, not reasonably be assumed to be exogenous. $\Delta Tax_{c(j),t}$ measures the carbon tax change from *t-1* to t at country-level, i.e. $\Delta Tax_{c(j),t} \equiv Tax_{c(j),t} - Tax_{c(j),t-1}$.

$X_{j,t-1}$ is a vector of firm controls at time *t-1*, which contains firm size (measured by *log(Assets)*), firm age (*young*, which is a dummy equal to 1 if firm age is less than five years and zero otherwise), and profitability (measured by *EBIT/Revenues*). These are key determinants of emission growth. [7] Across all specifications, we include country fixed effects (χ_c) to absorb time-invariant cross-country differences in carbon-tax levels, as documented in Section A.1. The baseline specification further includes industry-by-year fixed effects ($\delta_i \times \tau_t$), such that we compare more and less financially constrained firms within the same sector and year.[8] Thereby, our main specification controls for sector-specific technological trends, demand shocks, and common changes in emissions. Standard errors are clustered at the country level, corresponding to the level of variation in the carbon tax.

The coefficient of interest is β_1 on the interaction term $D2D_{j,t-1} \times \Delta Tax_{c(j),t}$, capturing how firms' credit constraints mediate the pass-through of carbon taxes to emissions growth. The identifying assumption is that the differential change in emissions of constrained and unconstrained in *untreated* countries provides a valid counterfactual for the differential response in *treated* countries in the absence of a change in carbon taxes. Importantly, we do not have to assume that the change in climate policy is exogenous with respect to aggregate credit constraints, but only require that changes in climate policy are not endogenous with respect to differences between the treatment and control group. In Appendix B.1, we provide further evidence that tight aggregate credit constraints do not predict climate policy changes at the country level.

2.3 Empirical Results

The main results are reported in Table 1. Firms with tighter credit constraints, i.e., a lower distance-to-default, exhibit a smaller decline in emissions after an emission tax increase. The standard deviation of the distance-to-default equals 5.50 in the full sample and 5.48

[7] As a robustness check, we also estimate specifications with extended control variable sets including capital intensity (*PPE/Assets*), liquidity (*CashFlow/Assets*), and investment (*CapEx/Assets*), reported in Table B.3, results are qualitatively unchanged.

[8] As a robustness check, we also estimate specifications with only time fixed effects (τ_t); results are qualitatively unchanged.

in the manufacturing sector, see Table A.2. Hence, a one–standard-deviation increase in distance-to-default is associated with a 1.1 percentage point emission reduction in the full sample (Column (1)) and a 2.75 percentage point reduction in the manufacturing sector (Column (2)). These effects are economically significant relative to the overall variation in firm-level emissions growth, which has a standard deviation of 0.91 percentage points in the full sample and 0.87 percentage points in the manufacturing sector, see Table A.1. The results in the manufacturing sector are particularly informative, as manufacturing firms dominate the sample (Table A.1), and because the firms in the model described in Section 3 are best interpreted as manufacturing firms. The effect of tighter credit constraints on emissions growth is around 2.5 times stronger in the manufacturing sector.

	(1) $\Delta \log(\text{Emi})_{j,t}$	(2) $\Delta \log(\text{Emi})_{j,t}$
$D2D_{j,t-1} \times \Delta Tax_{c(j),t}$	-0.002**	-0.005***
	(0.001)	(0.002)
$D2D_{j,t-1}$	0.005**	0.006***
	(0.002)	(0.002)
$\Delta Tax_{c(j),t}$	-0.017	0.018
	(0.011)	(0.014)
$\log(Assets)_{j,t-1}$	-0.022*	-0.034**
	(0.011)	(0.016)
$Young_{j,t-1}$	-0.021***	-0.019**
	(0.006)	(0.009)
$EBIT/Revenues_{j,t-1}$	-0.096***	-0.063***
	(0.020)	(0.020)
Constant	0.185*	0.287**
	(0.096)	(0.140)
Observations	40,109	23,984
R-squared	0.024	0.110
Industry-by-year FE	2-digit SIC	4-digit SIC
Country FE	YES	YES
Sectors	All	Manuf

Table 1: Carbon Taxes and Credit Constraints. This table reports the results of estimating Equation (1). Column (1) refers to the full sample, while column (2) to only manufacturing sector. Regressions are estimated at the firm-year level. $D2D_{j,t-1}$ is distance-to-default at $t-1$. $\Delta\text{Tax}_{c,t}$ is the change in the country-level carbon tax from $t-1$ to t. $D2D_{j,t-1} \times \Delta\text{Tax}_{c,t}$ is the interaction between $D2D_{j,t-1}$ and $\Delta\text{Tax}_{c,t}$. Controls are firm size ($\log(\text{Assets})_{j,t-1}$), age ($\text{Young}_{j,t-1}$), and profitability ($\text{EBIT/Revenues}_{j,t-1}$), all lagged by one year. We include country fixed effects and industry-by-year fixed effects in all specifications; industries are 2-digit in column (1) and 4-digit SIC in column (2). Standard errors (in parentheses) are clustered at the country level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The coefficient on $D2D_{j,t-1}$ in the second row indicates that firms with a high distance-to-default generally experience a larger growth of emissions. This is not surprising since less constrained firms, i.e. firms with high distance-to-default, tend to invest more and expand their business. Such firms have higher revenue growth, which is closely correlated with emissions (Zhang, 2024). The coefficients on firm-specific control variables are significant across almost all specifications. Emissions grow more slowly in large firms, which tend to be closer to their optimal size and hence have lower revenue and emissions growth. Firms that are very *Young*, i.e., less than five year olds and more profitable firms, with a high *EBIT/Revenues*, appear to be generally less emission-intensive.

In Appendix B.2, we augment our baseline specification by interacting these firm-level controls with the carbon tax shock to ensure that β_1 does not pick up a differential emission reduction response by firm size, age, and profitability, which are naturally correlated with emission intensities.[9] We additionally estimate specifications with extended control sets including capital intensity (*PPE/Assets*), liquidity (*Cashflow/Assets*), and investment (*CapEx/Assets*) in Section B.3. In all these extensions, the coefficient β_1 remains negatively significant, and its size even slightly increases.

Implicitly, our baseline specification assumes that firms respond to the climate policy in their primary location. Since the firms in our sample are large, it is reasonable to assume that many of them are multinational firms. These firms have the opportunity to shift carbon-intensive activities abroad in response to climate policy shocks in their headquarters (see also Bartram, Hou, and Kim, 2022). To alleviate this concern, we show in Section B.4 that our results are also robust to restricting the firm sample to the utilities sector. Such firms are predominantly active in a single country and, thus, less susceptible to shifting operations abroad.

Lastly, we classify firms as constrained and unconstrained based on the sector-specific median distance-to-default. We then estimate how emission growth responds to an carbon tax increase. Appendix B.5 provides details on the empirical specification, while Table B.5 reports the estimates. Emissions growth responds more strongly to carbon tax shocks among firms with higher distance-to-default, i.e. where credit constraints are less relevant. The advantage of such a median split is that the results can be directly mapped into our model. Therefore, we use this alternative specification to quantitatively discipline our model, to which we turn next.

3 Model

Time is discrete and denoted by $t = 1, 2, ...$ and each period corresponds to one year. The model features a representative household, capital good firms, and two types of final goods

[9] Kim (2023), Lanteri and Rampini (2023), and Capelle et al. (2023) document that emission intensities vary across firm age and size. Younger and smaller firms might therefore respond differently to climate policy for other reasons than tight credit constraints.

and manufacturing firms, respectively. Emissions, climate policy, and financial frictions enter at the stage of manufacturing firms.

Manufacturing Firms: Technology There are two types of manufacturing firms that differ in their financing frictions but are otherwise identical. Both firms are managed by risk-neutral firm managers and distribute their dividends to the representative household. One firm type turns out to be *risk-free*, as its managers do not have an incentive to issue debt, and we denote its choice variables by the superscript *rf*. The other *risky* type is subject to credit frictions, and its decision variables are indicated without a superscript.

Both firm types use a production technology that is linear in capital (k_t and k_t^{rf} are subject to uninsurable idiosyncratic productivity shocks. The intermediate goods of both types z_t and z_t^{rf} are combined with labor into a homogeneous final good, described below. The idiosyncratic productivity shock is i.i.d. across firms and time, and follows a log-normal distribution with standard deviation ς. We normalize its mean to $-\frac{\varsigma^2}{2}$, which ensures that expected productivity equals one. During the production process, manufacturers generate emissions, which inflict a utility loss on households, described below. Emissions are proportional to production, consistent with empirical evidence presented in Zhang (2024). Furthermore, emissions are taxed at the rate τ_t, and manufacturers can reduce their emissions by operating a cleaner but less efficient technology.[10]

Following the E-DSGE literature (Heutel, 2012), we model such a technology choice as the option to mitigate the share a_t of their emissions. The cost of emission abatement per unit of output is given by $\frac{\alpha_0}{1+\alpha_1} a_t^{\alpha_1}$.

Safe Manufacturing Firms: Maximization Problem We first describe the maximization problem for risk-free firms, which is standard in the E-DSGE literature, and requires us to characterize two endogenous choice, the abatement effort a_t^{rf} and investment k_t^{rf}. For risk-free firms, emissions are given by $e_t^{rf} = (1 - a_t^{rf}) z_t^{rf}$. Since these firms do not issue debt, they are not subject to default risk, such that their dividends are given

[10] We do not allow firms to move their emission-intensive activities abroad, and we also assume that each firm's emissions are public knowledge. While these simplifications simplify the modeling of credit constraints and firm emissions, there is recent empirical and theoretical work along these lines: Bartram, Hou, and Kim (2022) show that financially constrained firms respond to a tightening of climate policy in California by shifting their production to other states. Similarly, Berg, Ma, and Streitz (2025) show that large public firms sell emission-intensive assets after the Paris Agreement to other firms that are at lower levels of public scrutiny. Cartellier, Tankov, and Zerbib (2025) endogeneize emission disclosures at the firm level and Frankovic and Kolb (2024) demonstrate that imperfect emission disclosure negatively affects the conduct of climate policy. Carbon leakage and imperfect emission disclosure further impair the efficacy of climate policy.

by

$$div_t^{rf} = p_t^{Z,rf} z_t^{rf} - (1 - a_t^{rf})\tau_t - \frac{\alpha_0}{1+\alpha_1}\left(a_t^{rf}\right)^{1+\alpha_1} - p_t^K i_t^{rf} \ . \tag{2}$$

Here, $p_t^{Z,rf}$ is the output price and p_t^K the price of investment goods, which we normalize to one in the quantitative analysis. Without default risk, we can directly integrate out the idiosyncratic productivity shock $z_t = k_t$, since m_t has a mean of one by assumption. The abatement effort a_t^{rf} that maximizes period t dividends is given by the familiar expression

$$a_t^{rf} = \left(\frac{\tau_t}{\alpha_0}\right)^{\frac{1}{\alpha_1}} \ , \tag{3}$$

The investment choice for risk-free firms solves the shareholder value maximization problem $\mathbb{E}_0[\sum \beta^t div_t^{rf}]$, subject to the law of motion for capital $i_t^{rf} = k_{t+1}^{rf} - (1-\delta_K)k_t^{rf}$

$$p_t^K = \mathbb{E}_t\left[\Lambda_{t,t+1}\left\{(1-\delta_K)p_{t+1}^K + p_{t+1}^{Z,rf} - (1-a_t^{rf})\tau_{t+1} - \frac{\alpha_0}{1+\alpha_1}\left(a_t^{rf}\right)^{1+\alpha_1}\right\}\right] \ , \tag{4}$$

where $p_{t+1}^{Z,rf}$ is the price of intermediate goods next period. Optimal investment equates the price of capital on the LHS with the value of undepreciated capital next period, the sales of output per unit of investment, reduced by the per-unit emission tax $(1-a_t^{rf})\tau_{t+1}$ and the per-unit abatement cost.

Risky Manufacturing Firms: Credit Frictions Risky manufacturing firms are managed on behalf of impatient firm owners, who have a subjective discount factor smaller than the household discount factor ($\widetilde{\beta} < \beta$). Firm managers prefer to front-load dividend payments, which gives rise to a borrowing motif. They finance their investment with defaultable long-term debt l_t and with equity, which is modeled as a transfer from households. As customary in the literature, a share χ of all outstanding debt matures each period. The repayment obligation coming into period t is given by χl_t, and a firm defaults if after-tax revenues are insufficient to cover the repayment obligation. The implicit assumption is that firms can not raise outside equity solely to repay their creditors. The threshold productivity level $\overline{m}_t$ below which a firm defaults is given by

$$\overline{m}_t \equiv \frac{\chi l_t}{(p_t^Z - \tau_t(1-a_t))k_t} \ . \tag{5}$$

In the event of a corporate default, all output is lost. It is helpful to introduce two definitions related to firm default. The expected profitability of a defaulting firm is denoted by $G(\overline{m}_{t+1}) \equiv \int_0^{\overline{m}_{t+1}} m dF(m)$ and the default probability by $F(\overline{m}_{t+1}) \equiv \int_0^{\overline{m}_{t+1}} dF(m)$. To maintain tractability, we follow Gomes, Jermann, and Schmid (2016) in assuming that a

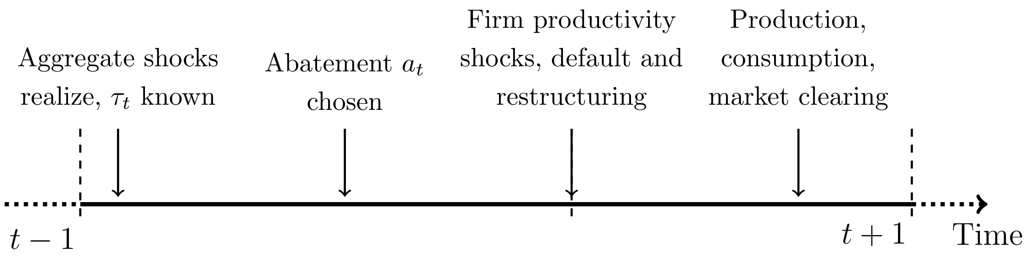


Figure 1: Within-period timing assumption.

defaulting firm is restructured immediately, such that it re-enters the debt market in the default period. This facilitates aggregation into a representative manufacturing firm.

Putting these things together, firm dividends in period t can be written as

$$\begin{aligned} div_t =& \mathbb{1}\{m_t > \overline{m}_t\} \cdot \Big(\big(p_t^Z - \tau_t(1-a_t) \big) z_t - \chi l_t \Big) \\ & - p_t^K i_t - \frac{\alpha_0}{1+\alpha_1} \big(a_t\big)^{1+\alpha_1} k_t + q(\overline{m}_{t+1}) \Big(l_{t+1} - (1-\chi) l_t \Big) \, . \end{aligned} \tag{6}$$

The first term reflects the after-tax production revenues $\widetilde{p}_t^Z z_t$, net of debt repayment χl_t, in the repayment case where the productivity draw m_t exceeds the default threshold $\overline{m}_t$. The second line of Equation (6) contains all investment and financing activity in period t that is relevant for period $t+1$. Due to the assumption of immediate restructuring, a firm can invest in capital, exert abatement effort, and change its net debt position $l_{t+1} - (1-\chi)l_t$ irrespective of its productivity draw. In addition to the choice variables of risk-free firms, risky firms also choose the debt level l_t, which in turn affects default risk $\overline{m}_{t+1}$ in future periods and, hence, debt prices $q(\overline{m}_{t+1})$ in the current period.

Banks have a minimal role in our model. They collect deposits d_{t+1} from households that promise to pay $(1+r_t)d_{t+1}$ units of the consumption good in the next period and invest the proceeds into corporate debt issued by risky manufacturing firms l_{t+1}. The (realized) per-unit payoff from holding corporate debt is denoted by $\mathcal{R}_t$. Following Cúrdia and Woodford (2011), we impose that banks have to repay their depositors in expectations: $(1+r_t)d_{t+1} = \mathbb{E}_t[\mathcal{R}_{t+1}]l_{t+1}$. Solving the representative banks' profit maximization problem yields the debt pricing condition:

$$q(\overline{m}_{t+1}) = \frac{1}{1+r_t} \mathbb{E}_t[\mathcal{R}_{t+1}] \, . \tag{7}$$

Risky Manufacturing Firms: Maximization Problem Similar to the case of risk-free firms, we can divide the maximization problem of risky firms into two stages. In the intra-period stage, we derive the optimal abatement choice a_t, which is chosen before idiosyncratic productivity shocks realize, and before capital k_t and debt l_t can be adjusted.

Due to the assumption of immediate restructuring after a potential default event, the relevant part of the expected period t dividends are

$$\mathbb{E}_{m_t}[div_t] = \big(1 - G(\overline{m}_t)\big)\big(p_t^Z - \tau_t(1-a_t)\big)k_t - \big(1 - F(\overline{m}_t)\big)\chi l_t - \frac{\alpha_0}{1+\alpha_1} a_t^{1+\alpha_1} k_t \ . \tag{8}$$

In Appendix C, we show that the optimal abatement effort is given by

$$a_t = \left(\frac{\big(1 - G(\overline{m}_t)\big)\tau_t}{\alpha_0}\right)^{\frac{1}{\alpha_1}} \ . \tag{9}$$

This expression collapses to the abatement effort of risk-free firms, equation (3), if there is no default risk. The expected productivity conditional on not defaulting $\big(1 - G(\overline{m}_t)\big)$ governs how firm credit frictions impair the transmission of climate policy to emission abatement. Conceptually, equation (9) also reveals that this impairment arises from frictions on the credit demand side. There is no friction associated with creditors of the firm, and the impairment operates entirely through firms' ex ante risk choice and ex post debt overhang problems. It follows immediately from equation (9) that the long run carbon tax consistent with full abatement τ^0 is given by $\frac{\alpha_0}{1-G(\overline{m})}$. We define the carbon tax level consistent with full abatement by unconstrained firms as $\tau^{0,rf} \equiv \alpha_0$.

In a second step, we derive firms' optimal investment and debt choices as the solution to an inter-temporal shareholder value maximization problem $\mathbb{E}_0[\sum \widetilde{\beta}^t div_t]$. Firms maximize profits, taking as given the loan price schedule, which is obtained from plugging the debt payoff into banks' debt pricing condition (6). The (per-unit) debt payoff can be written

$$\mathcal{R}_{t+1} = \chi\Big(1 - F(\overline{m}_{t+1})\Big) + (1-\chi)q(\overline{m}_{t+2}) \ . \tag{10}$$

The first term of Equation (10) reflects the expected payoff from the share χ of maturing debt. With probability $1 - F(\overline{m}_{t+1})$, the firm repays. The second term is the rollover share $(1-\chi)$ of outstanding debt, valued at next period's market price $q(\overline{m}_{t+2})$. After plugging in the law of motion for capital $i_t = k_{t+1} - (1-\delta_K)k_t$, we can reduce the firm maximization problem to a two-period consideration:

$$\begin{aligned}
\max_{k_{t+1}, l_{t+1}, \overline{m}_{t+1}} & -p_t^K k_{t+1} - \frac{\alpha_0}{1+\alpha_1}\big(a_{t+1}\big)^{1+\alpha_1} + q(\overline{m}_{t+1})\Big(l_{t+1} - (1-\chi)l_t\Big) + \\
& \mathbb{E}_t\Bigg[\widetilde{\Lambda}_{t,t+1} \cdot \Bigg\{\int_{\overline{m}_{t+1}}^{\infty} \big(p_{t+1}^Z - \tau_{t+1}(1-a_{t+1})\big)m_{t+1}k_{t+1} - \chi \cdot l_{t+1} dF(m_{t+1}) \\
& + p_{t+1}^K(1-\delta_K)k_{t+1} + q(\overline{m}_{t+2})\Big(l_{t+2} - (1-\chi)l_{t+1}\Big)\Bigg\}\Bigg] ,
\end{aligned}$$

subject to the default threshold (5), the debt pricing condition (7), and the debt payoff

(10). Solving the risky firm's maximization problem, we obtain the following first-order conditions

$$p_t^K - \mu_t \frac{\overline{m}_{t+1}}{k_{t+1}} = \mathbb{E}_t\left[\widetilde{\Lambda}_{t,t+1}\left\{(1-\delta_K)p_{t+1}^K + \big(1-G(\overline{m}_{t+1})\big)\widetilde{p}_{t+1}^Z\right\}\right] , \tag{11}$$

$$q(\overline{m}_{t+1}) - \mu_t \frac{\overline{m}_{t+1}}{l_{t+1}} = \mathbb{E}_t\left[\widetilde{\Lambda}_{t,t+1}\left\{\chi(1-F(\overline{m}_{t+1})) + (1-\chi)q(\overline{m}_{t+2})\right\}\right] , \tag{12}$$

$$\begin{aligned} &-\mu_t - q'(\overline{m}_{t+1})\Big(l_{t+1} - (1-\chi)l_t\Big) = \\ &\qquad \mathbb{E}_t\left[\widetilde{\Lambda}_{t,t+1}\left\{\Big(l_{t+2} - (1-\chi)l_{t+1}\Big)q'(\overline{m}_{t+2})\frac{\partial \overline{m}_{t+2}}{\partial \overline{m}_{t+1}}\right\}\right] . \end{aligned} \tag{13}$$

Here, μ_t denotes the Lagrange multiplier on the default threshold, equation (5). Similar to risk-free firms, the first-order condition for capital, equation (11), balances the cost of purchasing one unit of capital p_t^K with the expected after-tax revenue it generates in $t+1$, its resale value $(1-\delta_K)p_{t+1}^K$. For risky firms, the optimal capital choice also takes into account that more investment reduces next period's default threshold and hence raises current dividends, reflected by the expression $-\mu_t \frac{\overline{m}_{t+1}}{k_{t+1}}$ on the LHS. It also takes into account that firms do not always receive the payoff from their investment, reflected by the term $\big(1-G(\overline{m}_{t+1})\big)$ on the RHS.

The two additional first-order conditions are not present for risk-free firms. The corporate capital structure is determined according to equation (12). Issuing debt raises dividends in period t by $q(\overline{m}_{t+1})$ units, which has to equal the expected repayment obligation and a debt roll-over term in period $t+1$. As the LHS of equation (12) shows, the debt choice also takes into account how an additional unit of debt affects the default threshold. Lastly, equation (13) links the multiplier on the capital structure choice to the elasticity of the debt price $q'(\overline{m}_{t+1})$, Since debt is long-term, the capital structure choice takes into account that increasing the default risk today is also linked to next period's default risk through the policy function for capital structure $\overline{m}_{t+2}(\overline{m}_{t+1})$.

Final Good Firms There are two types of final good firms, which combine the intermediate goods produced by the manufacturing sector with labor into a homogeneous final good using a Cobb-Douglas technology. Denoting the output by the final good firms by $y_t = A_t z_t^{\theta} n_t^{1-\theta}$ and $y_t^{rf} = A_t (z_t^{rf})^{\theta}(n_t^{rf})^{1-\theta}$, solving each (static) maximization problem yields standard first-order conditions, which we report in Apppendix C.

Households The representative household has preferences

$$u(c_t, n_t, n_t^{rf}, \Delta\mathcal{T}_t) = \log(c_t) - \frac{\omega_N}{2}(n_t)^2 - \frac{\omega_N}{2}\big(n_t^{rf}\big)^2 - \gamma_T(\Delta\mathcal{T}_t)^2 , \tag{14}$$

where c_t is consumption, n_t is labor supply to constrained intermediate good firms, n_t^{rf} is labor supply to unconstrained intermediate good firms, and $\Delta\mathcal{T}_t$ is the global temperature increase since pre-industrial times, described in the next section. Writing damages associated with carbon dioxide emissions into households' utility function is a common assumption in the E-DSGE literature, see for example Acemoglu et al. (2012) and Barrage (2019).

Households save in the form of bank deposits that earn the real interest rate r_t. We denote their time-preference parameter by β, which pins down the (steady state) real interest rate in equilibrium. Households supply labor to final good firms. The real wage is denoted by w_t. Solving the household maximization problem yields standard intra- and inter-temporal optimality conditions, see Appendix C.

Market Clearing and Climate Block The carbon tax revenue in period t is given by $\tau_t(1-a_t)z_t + \tau_t(1-a_t^{rf})z_t^{rf}$ and is rebated to households in lump sum fashion, such that market clearing on the final and investment goods market requires

$$y_t + y_t^{rf} = c_t + \frac{\alpha_0}{1+\alpha_1}\Big((a_t)^{1+\alpha_1} + (a_t^{rf})^{1+\alpha_1}\Big) + I_t\Big(1 + \frac{\Psi_K}{2}\big(\frac{I_t}{I_{t-1}} - 1\big)^2\Big) + k_t G(\overline{m}_t) \ , \tag{15}$$

and

$$I_t = i_t + i_t^{rf} \ . \tag{16}$$

Total emissions in period t are given by $e_t = (1-a_t)z_t$, which accumulate into a stock of atmospheric carbon dioxide

$$E_t \equiv \delta_E E_{t-1} + e_t \ , \tag{17}$$

where δ_E is the decay rate of atmospheric carbon dioxide. Following the climate economics literature (Fernandez-Villaverde, Gillingham, and Scheidegger, 2024), the global temperature change relative to pre-industrial levels, $\Delta\mathcal{T}_t \equiv \mathcal{T}_t - \mathcal{T}_{1850}$, is positively related to cumulated emissions. This relationship can be approximated by

$$\Delta\mathcal{T}_t = \Psi_{CCR} \cdot E_t \ , \tag{18}$$

where Ψ_{CCR} is the cumulative carbon response of global temperatures (in degrees Celsius) to one GtCO2 emissions. Squared temperature increases relative to pre-industrial times directly reduce households' utility.

4 Quantitative Analysis

In this section, we present the calibration of our model and discuss its external validity by comparing it to our empirical findings. We then quantify the macroeconomic relevance of firm credit frictions for welfare, climate policy, and the net zero transition.

4.1 Calibration

The model is solved and simulated using a second-order approximation around the deterministic steady state. While most parameters affecting households and final good firms are taken from the real business cycle literature, we use parameters from the environmental economics literature to discipline the climate block. To calibrate parameters related to financial frictions, we use annual macro-financial data from the US over the period 1981 to 2013. We also provide a calibration to a longer data sample from 1951 to 2013, which is characterized by a lower leverage ratio and default rate. Cutting of the sample in 2013 and hence before the Paris agreement, our data arguably precede the implementation of ambitious climate policies. Hence, we set the carbon tax to zero when computing model-implied moments. It should be stressed that we do not aim for a calibration specific to the US, but rather a reasonable parameterization in the context of advanced (OECD) economies to keep our model comparable to the empirical results of Section 2.

Households Parameters governing household preferences are set to standard values. The annual time discount factor of households is set to $\beta = 0.99$, implying a real risk-free rate of 1% p.a. Setting the weight $\omega_N = 8$ ensures that labor supply equals $n^{SS} = 0.33$ in the deterministic steady state.

To back out the temperature loss parameter γ_T, we leverage recent empirical evidence by Nath, Ramey, and Klenow (2024), who use cross-country temperature and GDP data and find that an increase of global surface temperatures by 3.7 degrees Celsius in 2100 reduces GDP by 7-12%. The upper end of this range is also consistent with estimates by Bilal and Rossi-Hansberg (2023) exploiting spatial heterogeneity within the US. Hence, we assume that GDP losses in 2100 amount to 12%/3.7°=3.24% per degree Celsius of global warming without any further climate policy tightening. We simulate the model under constant carbon taxes with $T_0 = 2024$ as the start date, such that $E_{2024} = 2.400$ GtCO2 corresponds to the stock of emissions in the initial period. The loss parameter $\gamma_T = 0.0185$ is then implicitly defined through $u\big(c_{2100}, \Delta\mathcal{T}^2_{2100}\big) = u\big((1 - 0.0324 \cdot \Delta\mathcal{T}_{2100}) \cdot c_{2100}, 0\big)$. In Section 4.5, we discuss how changes to the temperature loss parameter affect our welfare results.

Parameter	Value	Source/Target
Households		
Household discount factor β	0.99	1% real risk-free rate
Labor disutility weight ω_N	8	Labor supply $n^{SS} = 0.33$
Temperature losses ω_T	0.0147	Nath, Ramey, and Klenow (2024)
Technology and Climate		
Cobb-Douglas coefficient θ	1/3	Capital share of production
Capital depreciation rate δ_K	0.08	In line with RBC literature
Abatement cost parameter α_0	0.033	Abatement/GDP ratio
Abatement cost parameter α_1	1.6	Heutel (2012)
Emission decay δ_E	0.99	1% decay of CO2 p.a.
Temperature parameter Ψ_{CCR}	0.58	Fernandez-Villaverde et al (2024)
Credit Frictions		
Firm owner discount factor $\widetilde{\beta}$	0.91	Corporate default rate
St. dev. firm productivity ς	0.3	Corporate leverage ratio
Debt maturity parameter χ	0.2	5-year average debt maturity
Shocks		
Persistence TFP ρ_A	0.8	In line with RBC literature
TFP shock st. dev. σ_A	0.005	In line with RBC literature

Table 2: Calibration.

Technology The next group of parameters affects technology and emissions. For both firm types, we set the capital depreciation rate to $\delta_K = 0.08$ and the Cobb-Douglas coefficient to $\theta = 1/3$. Exogenous TFP follows an AR(1) process in logs

$$\log(A_t) = \rho_A \log(A_{t-1}) + \sigma_A \epsilon_t^A \ . \tag{19}$$

We set $\rho_A = 0.8$ and $\sigma_A = 0.005$, which are standard values for business cycle models at annual frequencies. Following the E-DSGE literature, for example Heutel (2012), the abatement cost function is parameterized to be consistent with Nordhaus (2018). We set the curvature parameter to its standard value of $\alpha_1 = 1.6$. Setting the slope parameter to $\alpha_0 = 0.033$ delivers an abatement cost-to-GDP ratio of 1.3% under full abatement, in line with comparable smaller-scale E-DSGE models (Carattini, Melkadze, and Heutel, 2023). An carbon tax of 94\$/tCO2 achieves full abatement for unconstrained firms under this parameterization.[11] To map emissions into global temperature increases, we use a value

[11] We map the carbon tax in the model into a more easily interpretable price in \$/tCO2 using world GDP ($y^{world} = 105$ trillion USD in 2022, at PPP, see IMF, 2022) and world emissions ($e^{world} = 33$ GtCO2 in 2022). Absent carbon taxes, the model implies an output and emission level of $e^{model} = z^{model} = z + z^{rf} = 2.6$. The emission price in \$/tCO2 associated with a given tax τ_t is thus given by

of $\Psi_{CCR} = 0.58$ degrees Celsius per GtCO2 emissions, following Fernandez-Villaverde, Gillingham, and Scheidegger (2024).

Financial Frictions The last set of parameters relates to credit frictions, and they are calibrated using historical macro-financial data. Hence, we assume that the carbon tax is zero when computing model-implied moments. We set $\chi = 0.2$ to obtain an average debt maturity of five years. The standard deviation of the idiosyncratic productivity shock $\varsigma = 0.3$ and the firm owner discount factor $\widetilde{\beta} = 0.91$ that applies to risky firms are jointly calibrated to match the firm default frequency and firm leverage. We target a leverage ratio of 42%, which corresponds to the mean of the US non-financial business sector from 1981 to 2013. In the data, leverage is defined as all borrowing, divided by physical assets at book values and cash holdings (Gomes, Jermann, and Schmid, 2016). Leverage in the model is defined as $lev \equiv l/k$. We also target a corporate default rate of 4.5%, which corresponds to the average default rate of speculative grade US non-financial firms. Associating such firms with the risky/constrained firm type in our model appears to be a reasonable assumption. By contrast, the average default rate of investment grade-rated firms (Baa or higher) was below 0.1% and it is zero for safe/unconstrained firms in the model by construction. We also consider a robustness check where we target the leverage ratio and default rate over the longer time period 1951-2013. The parameterization is summarized in Table 2.

4.2 Model Fit

Table 3 presents the model's fit to the data. In the first panel, we show that the model closely matches the targeted corporate default rate and leverage ratio. The second panel demonstrates that model-implied macro-financial dynamics are consistent with the data. In particular, investment is considerably more volatile than investment and debt outstanding. The relative volatilities of leverage and default are somewhat smaller than in the data, since our model only features one shock in the production sector and no financial shocks at all. In line with the data, consumption, investment, and emissions are strongly pro-cyclical. While outstanding debt and leverage also exhibit substantial pro-cyclicality (Jungherr and Schott, 2022), corporate default risk is counter-cyclical (Kuehn and Schmid, 2014).

The last row of Table 3 demonstrates that our calibrated model implies a reasonable impairment of the pass-through of carbon taxes to the abatement effort. In the data, credit-constrained firms experience a 0.5 percentage point smaller emission reduction after

$p_t^e = \frac{y^{\text{world}}/z^{\text{model}}}{e^{\text{world}}/e^{\text{model}}} \cdot \tau_t = 2,798 \cdot \tau_t$.

Moment	**Model**	**Data**
Targeted Moments		
Corporate default rate (%)	4.5	4.5
Leverage (%)	42	42
Untargeted Moments		
Relative vol. consumption $\sigma(c)/\sigma(y)$	0.96	0.84
Relative vol. investment $\sigma(i)/\sigma(y)$	3.35	4.23
Relative vol. debt $\sigma(b)/\sigma(y)$	1.17	1.84
Relative vol. leverage $\sigma(lev)/\sigma(y)$	0.16	0.55
Relative vol. default $\sigma(F(\overline{m}))/\sigma(y)$	0.28	1.22
Correlation GDP-Emissions cor(e, y)	0.59	0.76
Correlation GDP-Consumption cor(c, y)	0.59	0.91
Correlation GDP-Investment cor(i, y)	0.48	0.89
Correlation GDP-Debt cor(y, l)	0.13	0.48
Correlation GDP-Leverage cor$(y, F(\overline{m}))$	0.27	0.23
Correlation GDP-Default cor$(y, F(\overline{m}))$	-0.25	-0.31
Effect of 10\$/tCO2 Tax Increase		
$\Delta\big(\log(e^{rf}) - \log(e)\big)$	-0.5%	-0.5%

Table 3: Model Fit: $\Delta\big(\log(e_t^{rf}) - \log(e)_t\big)$ refers to the difference between the (relative) emission change of unconstrained (rf) and constrained firms, expressed in percentage points. All model-implied moments are computed based on a second-order approximation around the deterministic steady state and are expressed in relative deviations from their steady state value. The second moments in the data are based on US data from 1951-2013, de-trended using an HP-filter with a smoothing parameter of 6.25. GDP, investment, corporate debt, and consumption are logged and expressed in real terms.

a 10$/tCO2 carbon tax hike. Mimicking our empirical strategy, we subject the economy to an unanticipated and permanent carbon tax increase, starting from an initial level of zero to 10$/tCO2. We solve for the transition dynamics non-linearly using Dynare (Adjemian et al., 2022). The model-implied difference in emission reduction amounts to 0.5%, which comes very close to our empirical findings. This further corroborates the external validity of our calibration, since we did not use this differential response to carbon taxes to calibrate financial frictions or the climate block. Furthermore, the model-implied difference is slightly smaller than our empirical findings. This suggests that our calibrated model provides a conservative assessment of the relevance of firm credit constraints for climate policy.

4.3 Climate Policy Implications

So far, we have demonstrated our model's capability to reconcile the empirically observed difference in the emission reduction of constrained and unconstrained firms in response to *unanticipated* carbon tax shocks. As a natural next step, we examine how credit constraints affect climate policy in the *long run*. To illustrate the role of firm credit constraints, we plot the abatement shares for risky (a_t) and risk-free (a_t^{rf}) firms for different tax levels in Figure 2. Comparing equation (9) and equation (3) immediately reveals that the term $(1-G(\overline{m}_{t+1}))$ governs the extent to which credit constraints impair the transmission of carbon taxes to emissions at the firm level. Intuitively, the impairment increases in the expected productivity conditional on defaulting $G(\overline{m}_{t+1})$, which is closely linked to the default probability $F(\overline{m}_{t+1})$. The baseline calibration implies that $G(\overline{m}_{t+1}) \approx 0.022$.

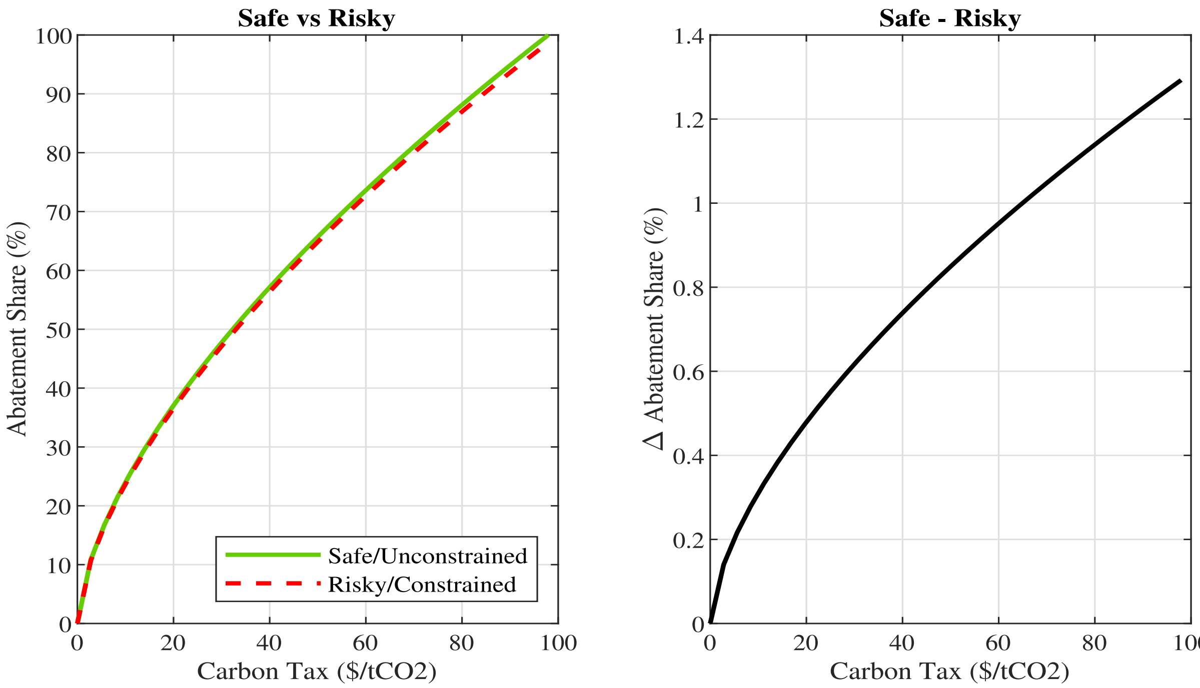


Figure 2: Credit Frictions and Optimal Abatement. The left panel displays optimal abatement effort by risky/constrained firms (dashed red line) and by safe/unconstrained firms (dotted green line) on the y-axis for different carbon tax levels on the x-axis. The solid black line in the right panel corresponds to the absolute difference (in percentage points) of the abatement shares.

The left panel of Figure 2 plots the long run abatement effort by safe/unconstrained firms (solid green line) and risky/constrained firms (dashed red line) for any given carbon tax level on the x-axis. While the abatement effort is an increasing and concave function of the tax for both firm types, it is generally larger for safe/unconstrained firms. The solid black line in the right panel plots their difference for different carbon tax levels. It stands out that the discrepancy increases with the carbon tax level. For example, the discrepancy is only around 0.35 percentage points for a tax level of 10\$/tCO2. This value is quite close to the differential emission reduction in response to a carbon tax *shock* (0.50 percentage points), but slightly smaller, since the dynamic response of emissions also captures that production by constrained firms declines more sharply. The discrepancy widens to around 1.3 percentage points for high carbon tax levels.

It is helpful to express this impairment using the carbon tax level necessary to induce full abatement of emissions. For risk-free/unconstrained firms, the tax consistent with emission abatement is given by $\tau_t = 0.033$, which translates into an emission price of 94.16\$/tCO2. For risky/constrained firms, the tax consistent with full abatement increases to 96.34\$/tCO2 (94.16/(1-0.022) \$/tCO2). While this difference appears small at first glance, it is actually sizable when compared to the current global coverage-weighted level of carbon pricing, which amounts to a little more than 5\$/tCO2.

4.4 Aggregate Implications and Welfare

In the previous sections, we have discussed the implications of credit constraints for the transmission of carbon taxes at the firm level, mimicking our empirical strategy. As a final step, we take our analysis to a macroeconomic level and study the transmission of carbon taxes to *aggregate* emissions in our baseline economy. We then examine the role of firm credit constraints by comparing aggregate emissions in the baseline economy to a counterfactual economy where risky/constrained firms choose the same abatement effort as safe/unconstrained firms.

We solve for the transition dynamics from zero to 10\$/tCO2 carbon tax non-linearly using Dynare (Adjemian et al., 2022). The results for our baseline economy are represented by red lines in Figure 3. Consistent with empirical evidence by Colmer et al. (2024), even a relatively modest tax hike substantially raises the abatement effort (middle left panel) and reduces emissions (bottom left panel). The reason is that abating emissions is very cheap if the tax level is initially zero or very low. Emissions decline by a little more (around 27%) than the abatement share (around 25%) as emission taxes also reduce investment and output. The top right panel demonstrates that the corporate default rate increases substantially on impact, consistent with empirical evidence provided by Berthold et al. (2023). Intuitively, firms' indebtedness becomes unsustainable if a larger share of revenues is spent on abatement and the emission tax bill. Once firms can adjust their risk-taking and debt issuance, the default rate quickly reverts to its initial level of 4.5%. The reason is that the carbon tax does not change firms' incentives to finance their investment with debt or equity in the long run.

We compare this economy to a counterfactual economy, in which risky/constrained firms' abatement effort is unaffected by financial frictions. Formally, we set $a_t = a_t^{rf} = \left(\frac{\tau_t}{\alpha_0}\right)^{\frac{1}{\alpha_1}}$. This is reflected by the dotted green lines in Figure 3. This is conceptually different from comparing the abatement effort of risky/constrained and safe/unconstrained firms in the same economy. While such a comparison within the same economy is more closely aligned with our empirical strategy and therefore useful for our calibration, it would omit important general equilibrium effects.

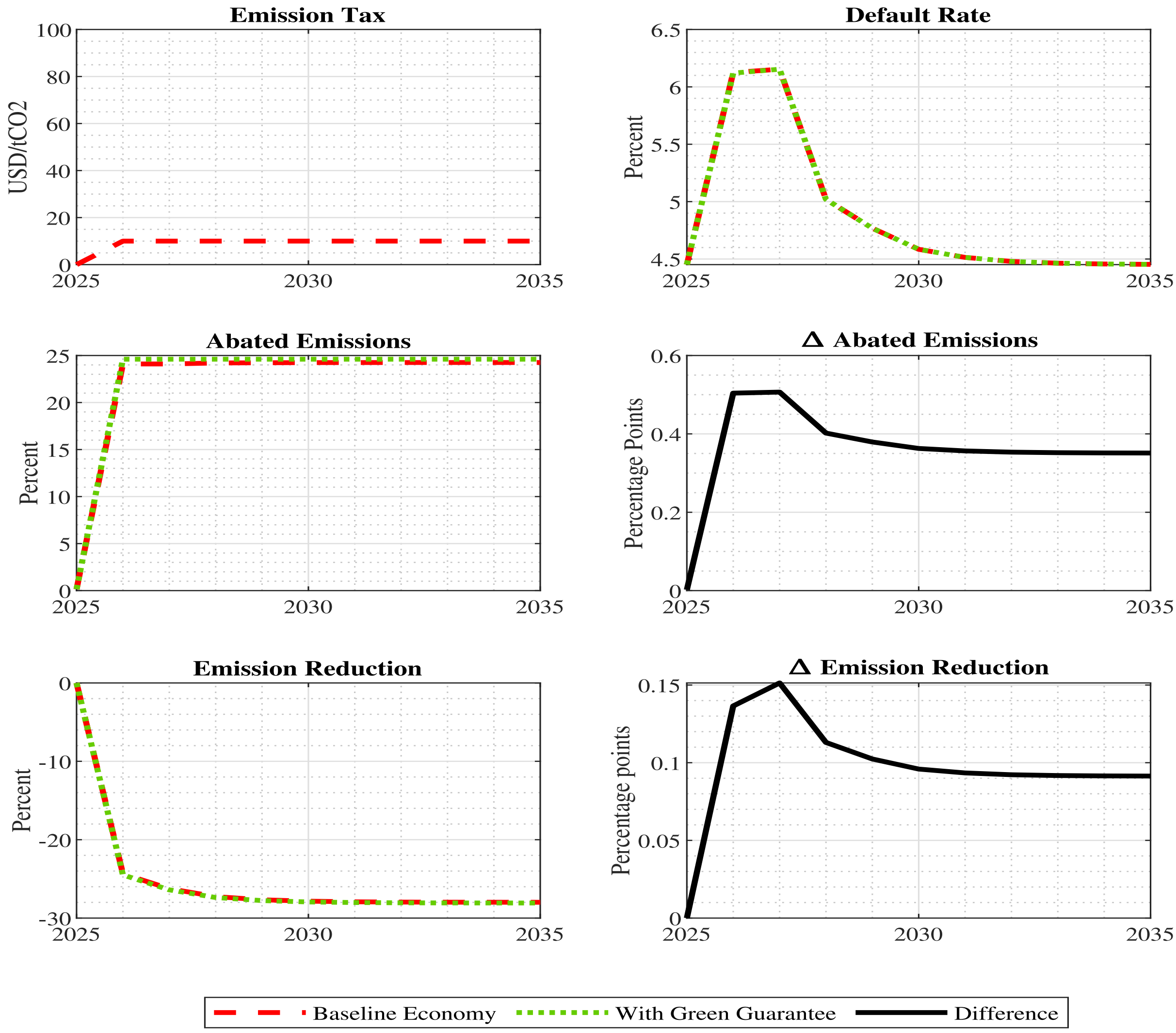


Figure 3: Credit Constraints and the Aggregate Transmission of Carbon Taxes. This figure displays the dynamic effects of an unanticipated and permanent carbon tax shock of 10\$/tCO2 (top left). The top right shows the default rate for risky/constrained firms in both economies. The share of abated emissions for the baseline economy (dashed red line) and the counterfactual without credit constraints/a green credit guarantee scheme (dotted green line) is shown in the middle left panel, and their difference is shown in the middle right panel. The bottom left shows the emission reductions for both economies, while the bottom right shows their difference, relative to an economy without any climate policy.

Since we are interested in the relevance of credit frictions for climate policy outcomes, we do not remove credit frictions entirely. If we were to remove the credit friction by setting their discount factor to β, previously risky/constrained firms would also invest and emit more, and we would not be able to isolate the role of credit constraints for emission reduction. Hence, the impatient firm type still defaults and, consequently, invests less in this counterfactual economy. As the top right panel shows, the default rate of risky/constrained firms in the counterfactual economy rises by almost two percentage points on impact, identical to the baseline economy. However, risky/constrained firms' abatement effort is almost 0.5 percentage points larger. By contrast, the abatement effort of safe/unconstrained firms is identical in both economies, which eventually translates into a 0.1% larger *aggregate* emission reduction in the counterfactual economy. The reason for this smaller aggregate effect is that only half the firms are constrained, and these firms are generally smaller than safe/unconstrained firms, which makes them relatively less important for aggregate emissions.

Welfare and Green Credit Guarantee Schemes Lastly, we assess how much firm credit constraints affect the welfare gain from raising carbon taxes from zero to 10\$/tCO2. Such a policy increases welfare by around 1.6% in consumption equivalents in the baseline model, in which half of the firms are credit-constrained. To isolate the role of credit constraints for the welfare gain of carbon taxes, we again consider the counterfactual model in which the abatement efforts of constrained and unconstrained firms coincide. The welfare loss of firm credit constraints amounts to slightly less than 0.01% in consumption equivalents.

It is worth noting that this abatement effort can be achieved by supporting risky firms with a state-dependent transfer, financed by lump sum taxes. Specifically, the owners of risky firms have to receive τ_t per unit of emissions $a_t k_t$ in the event of a default. The expected transfer is then given by $G(\overline{m}_{t+1})\tau_t a_t k_t$ and increases the payoff from abatement investment so that it exactly erases the negative effects of default risk and debt overhang. Importantly, the subsidy has to increase in default risk and the carbon tax level in order to induce risky/constrained firms to choose the same abatement level as safe/unconstrained firms. In practice, such a policy can be interpreted as a green credit guarantee scheme, as firms remain risky but are incentivized to invest in emission mitigation. Under this interpretation, such a green credit guarantee scheme would induce a small welfare gain, which amounts to around one percent of the total welfare gain of the carbon tax increase. The effect size is consistent with the welfare gains of other green financial policies studied in the literature, see for example Giovanardi et al. (2023) for green collateral policy and Carattini, Melkadze, and Heutel (2023) for green macroprudential policy.

4.5 Comparative Statics

In this section, we perform a number of comparative statics exercises with respect to key model parameters. The results are collected in Table 4. While column (1) represents the baseline calibration, column (2) presents the results from calibrating financial frictions to the full US sample from 1951 to 2013. Over the extended sample, the leverage ratio (32%) and the default rate of US speculative grade firms (2.8%) are somewhat lower. We match these moments by raising the firm discount factor to $\widetilde{\beta} = 0.94$.

In the third column, we increase the abatement cost slope parameter from $\alpha_0 = 0.033$ to

	(1) Baseline	(2) High $\widetilde{\beta}$	(3) High α_1	(4) Low γ_T	(5) Pers. a
Panel A: 10\$/tCO2 Shock					
$\Delta\big(\log(e_t^{rf}) - \log(e_t)\big)$ (%)	-0.5	-0.25	-0.35	-0.5	-0.04
$V^{constr} - V^{uncon}$ (%)	-0.005	-0.003	-0.005	-0.002	-0.005
$V^{constr} - V^{notax}$ (%)	+1.62	+1.62	+1.40	+0.79	1.58
Panel B: Net zero tax					
Unconstrained (\$/tC02)	94.16	94.16	128.40	94.16	94.16
Constrained (\$/tC02)	96.34	95.30	131.38	96.34	96.34

Table 4: Comparative Statics. Panel A reports the differential response of emissions by constrained and unconstrained firms $\Delta\big(\log(e_t^{rf})-\log(e_t)\big)$ to a 10\$/tCO2 carbon tax shock. The second row compares welfare in the baseline economy V^{uncon} to the counterfactual where credit constraints do not affect the abatement effort V^{constr}, which corresponds to the green credit guarantee scheme. The third row compares welfare in the baseline economy V^{uncon} to the economy without any climate policy V^{notax}. The welfare effects are expressed in consumption equivalents. The net zero tax levels for constrained and unconstrained firms are given by $\tau^0 = \frac{\alpha_1}{1-G(\overline{m})}$ and $\tau^{0,rf} = \alpha_1$, respectively, and converted into \$/tCO2 according to $p_t^e = \frac{y^{\text{world}}/y^{\text{model}}}{e^{\text{world}}/e^{\text{model}}} \cdot \tau_t = 2,798 \cdot \tau_t$.

$\alpha_0 = 0.045$, so that 1.7% of GDP is spent on abatement under net zero taxes. This value is in the upper range of abatement costs used in small-scale E-DSGE models and implies that raising carbon taxes from zero to 10\$/tCO2 has a generally smaller impact on emission abatement. Hence, the difference between risky/constrained and safe/unconstrained firms amounts to merely 0.35 percentage points. However, the welfare loss of credit constraints is almost identical to the baseline calibration. Due to the higher abatement cost, the carbon tax that is consistent with full abatement by safe/unconstrained firms is substantially larger (128.40\$/tCO2), while it rises to 131.38\$/tCO2 for risky/constrained firms.

The fourth column refers to the case with a lower damage parameter $\gamma_T = 0.0106$ that is consistent with the lower bound on the empirical estimates reported in Nath, Ramey, and Klenow (2024). Since emission damages enter households' utility function additively, this does not affect the interplay between firm credit frictions and climate policy. While lowering the damage parameter γ_T leaves the transmission of carbon taxes to emissions unaltered, it reduces the welfare relevance of this impairment. Specifically, welfare is merely 0.002% smaller if carbon taxes are raised from zero to 10\$/tCO2, i.e., the welfare relevance declines by around 50%. The overall welfare gain of raising carbon taxes is also halved to approximately 0.8% in consumption equivalents.

Finally, we consider an extension in which abatement is modelled as an investment good rather than a period-by-period choice. Here, a_t represents the stock of abatement goods, which follows the law-of-motion

$$a_t = (1-\delta_a)a_{t-1} + i_t^a \quad \text{and} \quad a_t^{rf} = (1-\delta_a)a_{t-1}^{rf} + i_t^{a,rf} \ . \tag{20}$$

We assume that firms choose abatement investment i_t^a and $i_t^{a,rf}$, respectively. Period dividends of constrained/risky forms then become

$$\mathbb{E}_{m_t}[div_t] = \big(1-G(\overline{m}_t)\big)\big(p_t^Z - \tau_t(1-a_t)\big)k_t - \big(1-F(\overline{m}_t)\big)\chi l_t - \frac{\alpha_0}{1+\alpha_1}(i_t^a)^{1+\alpha_1}k_t \ , \tag{21}$$

such that the first-order condition for abatement investment i_t^a reads

$$i_t^a = \left(\frac{\big(1-G(\overline{m}_t)\big)\tau_t}{\alpha_0}\right)^{\frac{1}{\alpha_1}} \ . \tag{22}$$

For the unconstrained/safe firm, this condition reduces to $i_t^{a,rf} = (\frac{\tau_t}{\alpha_0})^{\frac{1}{\alpha_1}}$. Different from the baseline model, equation (22) equates the marginal cost of increasing the stock of abatement with its benefit. The first-order conditions for physical investment and debt issuance are not affected by this extension. We impose the same depreciation rate δ_a for abatement and physical capital and then recalibrate the abatement cost slope parameter to $\theta_1 = 1.88$. Under this re-parameterization of the model, full emission abatement requires an abatement cost-to-GDP ratio of 1.3%. Furthermore, the same carbon tax rate implements net zero emissions for safe/unconstrained and risky/unconstrained firms, since financial frictions and hence the term $1 - G(\overline{m}_{t+1})$ are unaffected as well.

Column (5) of Table 4 demonstrates that the impact of the carbon tax shock on emissions by safe/unconstrained and risky/constrained firms is substantially smaller than in the baseline. The reason for this response is that building the abatement stock takes several years in this extension. The discrepancy in the emission reduction of both firm types widens considerably over time and is identical in the longer run. We show this graphically in Figure C.1 in Appendix C. Thus, the welfare gain of raising carbon taxes is slightly smaller, but the relative importance of credit constraints is not affected.

5 Conclusion

In this paper, we demonstrate that firm credit constraints impair the efficacy of climate policies. Using a cross-country dataset of emissions and credit constraints of publicly traded firms, we show that firms with tight credit constraints, measured by their distance-to-default, experience a 0.5 percentage point smaller emission reduction than unconstrained firms in the same industry after a carbon tax increase of 10\$/tCO2. We incorporate this channel into a DSGE model with carbon taxes, emission abatement, and financial frictions. There are two firm types: risky/constrained and safe/unconstrained firms. Credit constraints arise endogenously for the risky firm type, which does not always repay its debt and consequently does not always receive the payoff from its abatement investment. Hence, risky/constrained firms choose a lower abatement effort than safe/unconstrained firms.

Conceptually, our model gives rise to a credit demand channel that is associated with firm default risk and the resulting debt overhang problems. This channel operates independently of frictions on the credit supply side, which are well understood in the sustainable banking literature. In our framework, constrained firms underinvest in abatement not because creditors are unable to extend loans, but because default risk reduces the expected payoff from abatement investment. Put differently, the effective carbon price faced by constrained firms is scaled by their expected productivity conditional on repayment.

We calibrate the model to match key characteristics of firm credit frictions and climate policy. Importantly, the model is able to reconcile the untargeted differential response of constrained and unconstrained firms to a carbon tax increase observed in the data. We use the model to show that the lower abatement effort by risky/constrained firms reduces the welfare gains of increasing carbon taxes from zero to 10\$/tCO2 by slightly less than 0.01% in consumption equivalents, approximately one percent of the total welfare gain. This modest welfare effect is consistent with the quantitatively limited effect of credit supply-side interventions for climate policy outcomes documented in the literature.

Our findings have implications for the design of climate policy. The calibrated model implies that the carbon tax associated with full abatement is around 96\$/tCO2 for constrained firms, compared to 94\$/tCO2 for their unconstrained peers. Alternatively, policies that address the negative effects of default risk and debt overhang on abatement investment may increase the effectiveness of carbon pricing. State-dependent transfers to risky firms ("green credit guarantees") target the credit demand-side friction directly by ensuring that firm-owners receive the payoff from abatement returns in default states. This restores incentives to invest in abatement and improves welfare. A comprehensive welfare assessment of green credit guarantee schemes, in particular their financing and institutional implementation through financial markets, is left for future research.

References

Accetturo, Antonio, Giorgia Barboni, Michele Cascarano, Emilia Garcia-Appenini, and Marco Tomasi (2023). "Credit Supply and Green Investment." Working Paper.

Acemoglu, Daron, Philippe Aghion, Leonardo Bursztyn, and David Hemous (2012). "The Environment and Directed Technical Change." *American Economic Review* 102(1), 131–166.

Adjemian, Stephane et al. (2022). *Dynare: Reference Manual, Version 5*. Dynare Working Papers 80. CEPREMAP.

Almeida, Heitor, Murillo Campello, and Michael Weisbach (2004). "The Cash Flow Sensitivity of Cash." *Journal of Finance* 59(4), 1777–1804.

Bams, Denis, and Bram van der Kroft (2025). "Tilting the Wrong Firms? Sustainable Investing in Transitioning Firms." Working Paper.

Barrage, Lint (2019). "Optimal Dynamic Carbon Taxes in a Climate–Economy Model with Distortionary Fiscal Policy." *Review of Economic Studies*.

Bartram, Söhnke M., Kewei Hou, and Sehoon Kim (2022). "Real Effects of Climate Policy: Financial Constraints and Spillovers." *Journal of Financial Economics* 143(2), 668–696.

Bellon, Aymeric, and Yasser Boualam (2024). "Pollution-Shifting vs. Downscaling: How Financial Distress Affects the Green Transition." Working Paper.

Berg, Tobias, Robin Döttling, Xander Hut, and Wolf Wagner (2025). "Do Voluntary Initiatives Make Loans Greener? Evidence from Project Finance." Working Paper.

Berg, Tobias, Lin Ma, and Daniel Streitz (2025). "Out of Sight, Out of Mind: Divestments and the Global Reallocation of Pollutive Assets." Working Paper.

Berthold, Brendan, Ambrogio Cesa-Bianchi, Federico Di Pace, and Alex Haberis (2023). "The Heterogeneous Effects of Carbon Pricing: Macro and Micro Evidence." Working Paper.

Bilal, Adrien, and Esteban Rossi-Hansberg (2023). "Anticipating Climate Change Across the United States." Working Paper.

Campiglio, Emanuele, Alessandro Spiganti, and Anthony Wiskich (2024). "Clean Innovation and Heterogeneous Financing Costs." *Journal of Environmental Economics and Management* 128(103071). Working Paper.

Capelle, Damien, Divya Kirti, Nicola Pierri, and Germán Villegas Bauer (2023). "Mitigating Climate Change at the Firm Level: Mind the Laggards." Working Paper.

Carattini, Stefano, Givi Melkadze, and Garth Heutel (2023). "Climate Policy, Financial Frictions, and Transition Risk." *Review of Economic Dynamics* 51, 778–794.

Cartellier, Fanny, Peter Tankov, and Olivier Zerbib (2025). "Can Investors Curb Greenwashing?" *Journal of Economic Dynamics and Control* 180(105195). Working Paper.

Colmer, Jonathan, Ralf Martin, Mirabelle Muuls, and Ulrich Wagner (2024). "Does Pricing Carbon Mitigate Climate Change? Firm-Level Evidence from the European Union Emissions Trading System." *Review of Economic Studies.*

Covi, Giovanni, Maren Froemel, Dennis Reinhardt, and Nora Wegner (2025). "Climate Policy and Banks' Portfolio Allocation." Working Paper.

Cúrdia, Vasco, and Michael Woodford (2011). "The Central Bank Balance Sheet as an Instrument of Monetary Policy." *Journal of Monetary Economics* 58(1), 54–79.

De Haas, Ralph, Ralf Martin, Mirabelle Muuls, and Helena Schweiger (2024). "Managerial and Financial Barriers to the Green Transition." *Management Science* 71(4), 2751–3636.

De Haas, Ralph, and Alexander Popov (2022). "Finance and Green Growth." *Economic Journal* 133(650), 637–668.

Döttling, Robin, and Adrian Lam (2024). "Monetary Policy, Carbon Transition Risk, and Firm Valuation." Working Paper.

Döttling, Robin, and Magdalena Rola-Janicka (2025). "Too Levered for Pigou? A Model of Environmental and Financial Regulation." *Journal of Financial Economics* 172(104105).

Fang, Min, Po-Hsuan Hsu, and Chi-Yang Tsou (2023). "Pollution Abatement Investment under Financial Frictions and Policy Uncertainty." Working Paper.

Farre-Mensa, Joan, and Alexander Ljungqvist (2015). "Do Measures of Financial Constraints Measure Financial Constraints?" *Review of Financial Studies* 29(2), 271–308.

Fernandez-Villaverde, Jesus, Kenneth Gillingham, and Simon Scheidegger (2024). "Climate Change through the Lens of Macroeconomic Modeling." Working Paper.

Frankovic, Ivan, and Benedikt Kolb (2024). "The Role of Emission Disclosure for the Low-Carbon Transition." *European Economic Review* 167, 104792.

Gilchrist, Simon, and Egon Zakrajšek (2012). "Credit Spreads and Business Cycle Fluctuations." *American Economic Review* 102(4), 1692–1720.

Giovanardi, Francesco, and Matthias Kaldorf (2025). "Climate Change and the Macroeconomics of Bank Capital Regulation." Working Paper.

Giovanardi, Francesco, Matthias Kaldorf, Lucas Radke, and Florian Wicknig (2023). "The Preferential Treatment of Green Bonds." *Review of Economic Dynamics* 51, 657–676.

Goetz, Martin (2019). "Financing Conditions and Toxic Emissions." Working Paper.

Gomes, João, Urban Jermann, and Lukas Schmid (2016). "Sticky Leverage." *American Economic Review* 106(12), 3800–3828.

Haas, Christian, and Karol Kempa (2023). "Low-Carbon Investment and Credit Rationing." *Environmental and Resource Economics.*

Hale, Galina, Brigid Meisenbacher, Rami Najjar, and Fernanda Nechio (2025). "Industrial Composition of Syndicated Loans and Banks' Climate Commitments." Working Paper.

Heider, Florian, and Roman Inderst (2022). "A Corporate Finance Perspective on Environmental Policy." Working Paper.

Heutel, Garth (2012). "How Should Environmental Policy Respond to Business Cycles? Optimal Policy under Persistent Productivity Shocks." *Review of Economic Dynamics* 15(2), 244–264.

IMF (2022). *World Economic Outlook*. Tech. rep. International Monetary Fund.

Iovino, Luigi, Thorsten Martin, and Julien Sauvagnat (2023). "Corporate taxation and carbon emissions." Working Paper.

Jungherr, Joachim, and Immo Schott (2022). "Slow Debt, Deep Recessions." *American Economic Journal: Macroeconomics* 14(1), 224–259.

Kacperczyk, Marcin, and Jose-Luis Peydro (2022). "Carbon Emissions and the Bank Lending Channel." Working Paper.

Kaplan, Steven, and Luigi Zingales (1997). "Do Investment-Cash Flow Sensitivities Provide Useful Measures of Financing Constraints?" *Quarterly Journal of Economics* 112(1), 169–215.

Kim, Seho (2023). "Optimal Carbon Taxes and Misallocation across Heterogeneous Firms." Working Paper.

Kruse, Tobias, Antoine Dechezleprêtre, Rudy Saffar, and Leo Robert (2022). "Measuring environmental policy stringency in OECD countries: An update of the OECD composite EPS indicator." OECD Economics Department Working Papers No. 1703.

Kuehn, Lars-Alexander, and Lukas Schmid (2014). "Investment-Based Corporate Bond Pricing." *Journal of Finance* 69(6), 2741–2776.

Lanteri, Andrea, and Adriano Rampini (2023). "Financing the Adoption of Clean Technology." Working Paper.

Merton, Robert (1974). "On the Pricing of Corporate Debt: The Risk Structure of Itnerest Rates." *Journal of Finance* 29, 449–70.

Myers, Stewart C. (1977). "Determinants of corporate borrowing." *Journal of Financial Economics* 5(2), 147–175.

Nath, Ishan, Valerie Ramey, and Peter Klenow (2024). "How Much Will Global Warming Cool Global Growth?" NBER Working Paper 32761.

Nordhaus, William (2018). "Evolution of Modeling of the Economics of Global Warming: Changes in the DICE model." *Climatic Change* 148(4), 623–640.

Schuldt, Hendrik, and Kai Lessmann (2023). "Financing the low-carbon transition: the impact of financial frictions on clean investment." *Macroeconomic Dynamics* 27(7), 1932–1971.

Whited, Toni M., and Guojun Wu (2006). "Financial Constraints Risk." *Review of Financial Studies* 19(2), 531–559.

Xu, Qiping, and Taehyun Kim (2021). "Financial Constraints and Corporate Environmental Policies." *Review of Financial Studies* 35(2), 576–635.

Zhang, Shaojun (2024). "Carbon Returns across the Globe." *Journal of Finance* 80(1), 615–645.

Online Appendix to "Do Firm Credit Constraints Impair Climate Policy?"

Matthias Kaldorf and Mengjie Shi

March 11, 2026

A Data

A.1 Carbon Taxation

Figure A.1 shows how the carbon tax varies over time for the largest 28 countries in our sample. We do not specifically show the data for Slovenia, Slovakia, Ireland, Czech Republic and Hungary since we have at most 100 firm $\times$ year observations in those countries. The majority of countries in our sample did not change their carbon taxation between 2012 and 2019. We observe carbon tax changes in Canada, Denmark, France, Japan, Portugal, Spain, Switzerland and the United Kingdom. Most of those changes are tax hikes, although our sample also includes three tax decreases. As Figure A.2 suggests, there is considerable cross-country heterogeneity in the level of carbon taxes in our sample. Some countries like Finland, Norway, Sweden and Switzerland have permanently high carbon tax levels in our sample period, countries like France, Japan, Denmark, Poland and the United Kingdom experience intermediate tax levels. The remaining countries have carbon tax index close to zero throughout the sample period. The vertical line indicates 2015, i.e., the year in which the Paris agreement was signed. There has been no clearly visible climate policy tightening after 2015, which suggests that our results are not driven by "clustered" policy changes shortly after 2015.

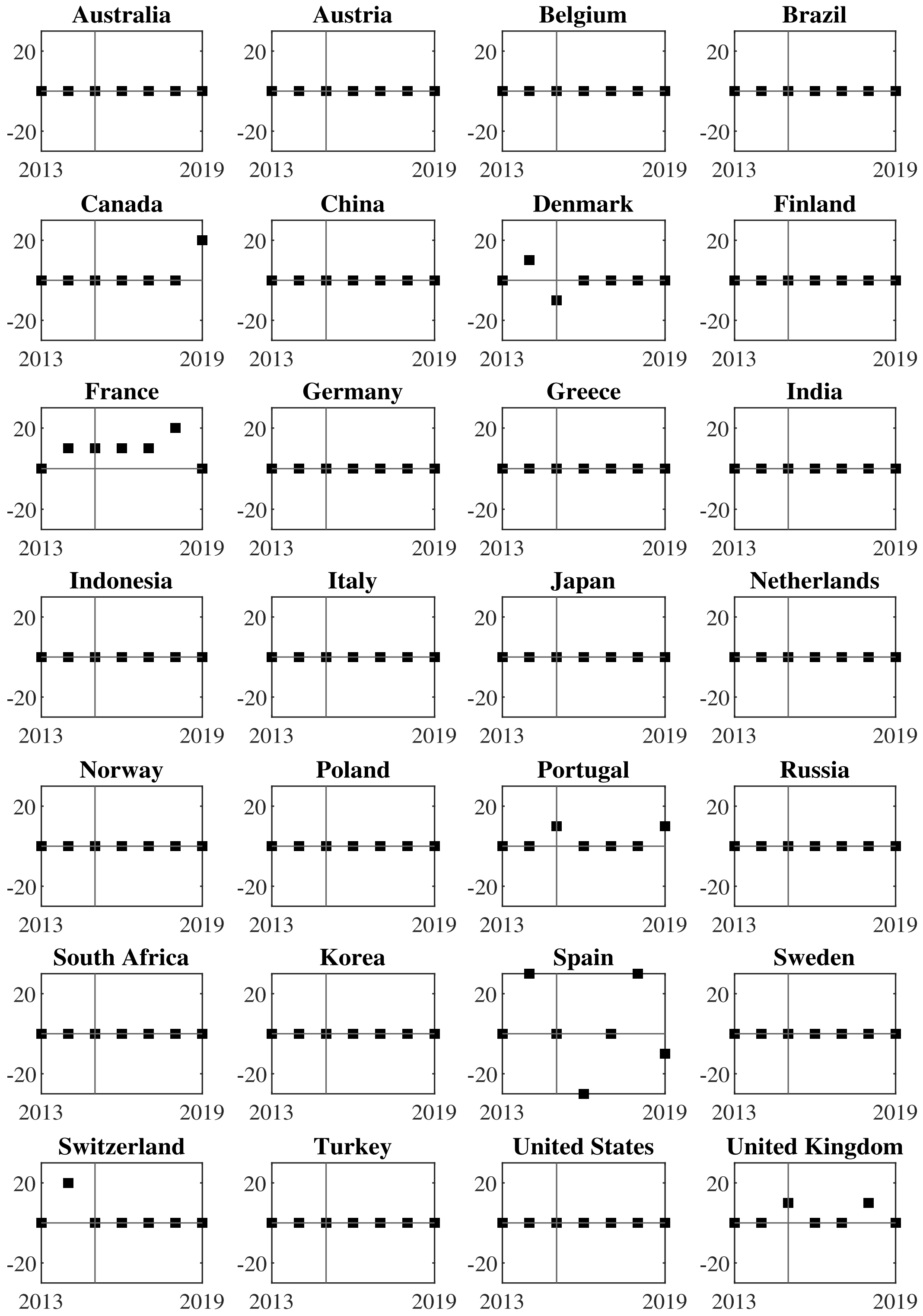


Figure A.1: Carbon Tax Shocks over Time ($/tCO2)

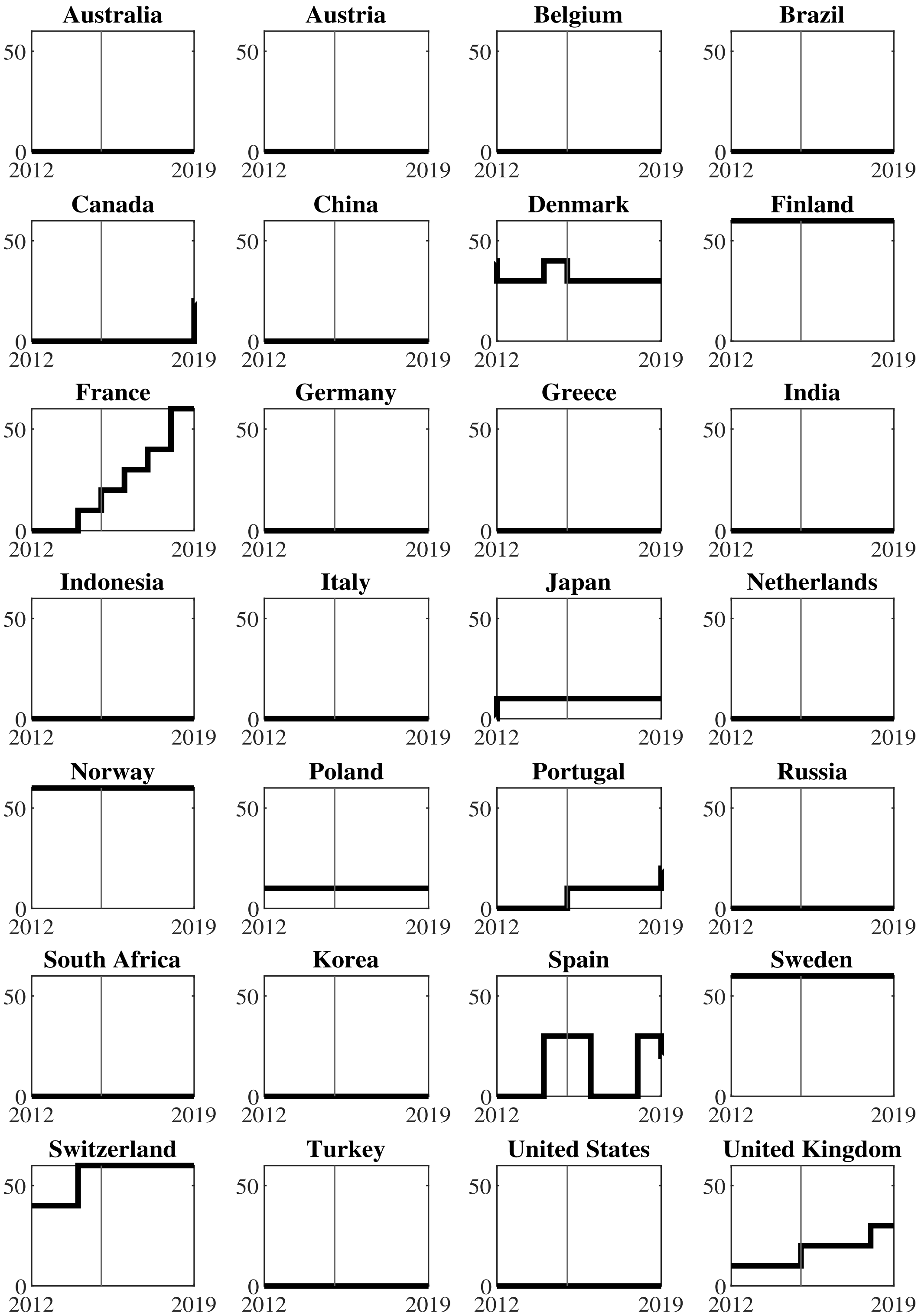


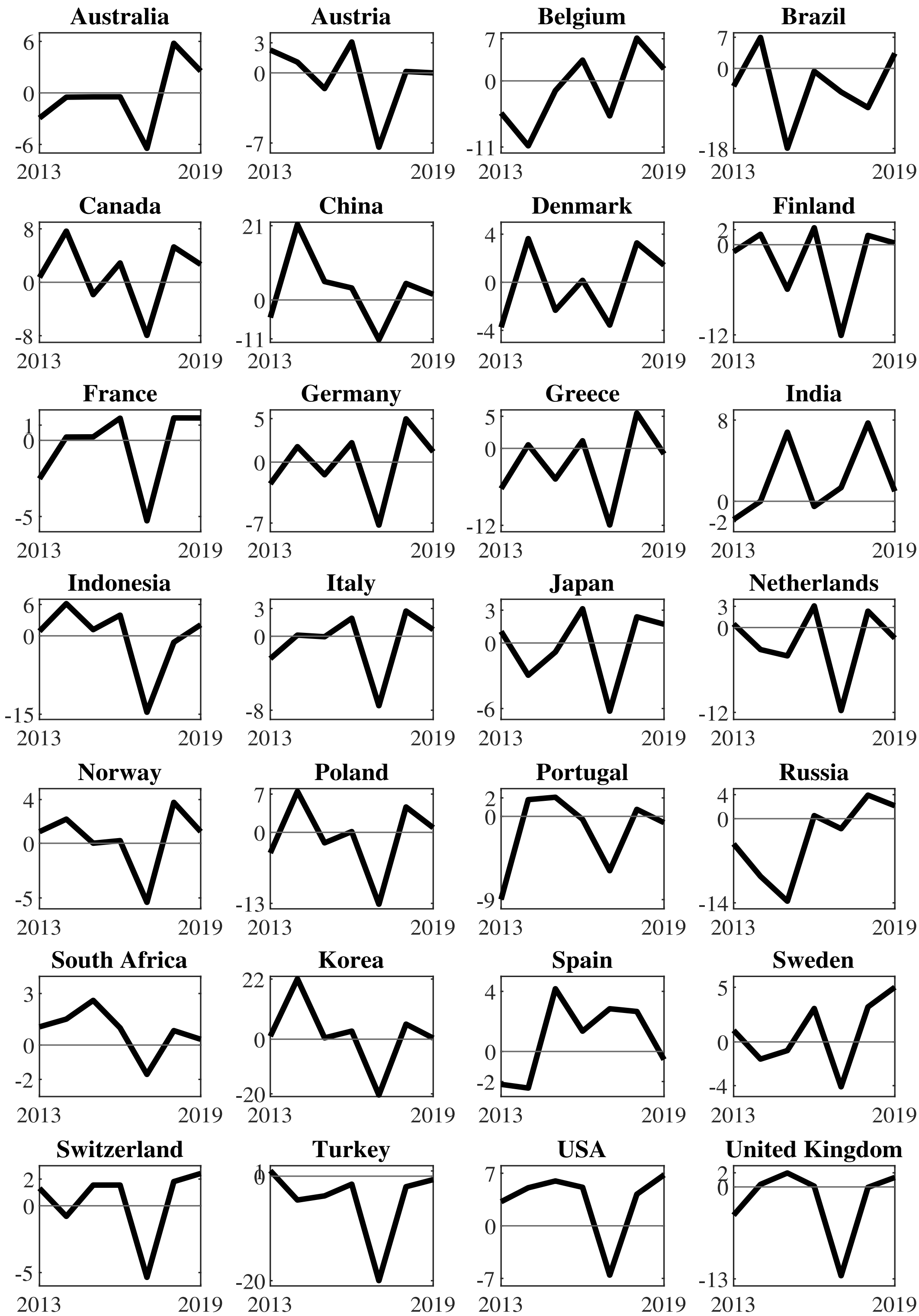
Figure A.2: Carbon Tax Level over Time ($/tCO2)

A.2 Emission Growth

Figure A.3 shows emission growth over time across countries. For each country and year, we compute the median emission growth (defined as its log difference) over all firms. In most countries, emission growth is positive in most years, with the notable exception being 2017. The pronounced emission reduction relative to 2016 can potentially be associated with the Paris Agreement, which was signed in December 2015 and came into force in 2016. Hence, we include industry $\times$ year fixed effects in our empirical specifications.

Table A.1 reports summary statistics for firm-level emissions growth by country and sector. Average emissions growth is close to zero across countries and industries, while the standard deviation is substantial, indicating pronounced variation in emissions dynamics across firms and over time. This heterogeneity underscores the importance of controlling for systematic differences in emissions trends driven by country-specific climate policies and sectoral characteristics. Accordingly, our baseline specification includes country fixed effects and industry-by-year fixed effects, ensuring that identification of the differential emissions response to carbon taxes operates within countries and within industries over time, rather than being driven by cross-country or cross-sectoral differences.

Figure A.3: Emission Growth over Time



	N	**Mean**	**SD**	**P25**	**P50**	**P75**
Panel A: Summary Statistics of Emission Growth across Countries						
Australia	2154	-0.49	1.15	-43.22	-0.46	43.59
Austria	241	-1.40	0.76	-25.27	-0.35	23.07
Belgium	380	2.33	1.00	-27.53	0.83	36.65
Brazil	778	2.91	1.21	-44.06	-2.07	45.32
Canada	1590	-0.05	0.93	-30.77	0.01	32.31
China	14080	-0.19	0.94	-40.29	0.00	40.26
Denmark	378	-2.45	0.88	-38.38	-3.40	31.57
Finland	637	-0.59	0.88	-31.42	-2.06	25.77
France	1797	-1.24	0.91	-28.42	0.07	24.54
Germany	1827	-0.44	0.83	-27.73	0.04	27.18
Greece	517	-4.34	0.94	-36.50	-0.87	32.49
India	4642	1.88	1.03	-33.52	0.66	38.05
Indonesia	1771	-1.36	0.99	-36.33	-0.67	32.49
Italy	723	-2.53	0.90	-27.39	-0.55	23.06
Japan	16545	0.03	0.77	-22.41	0.23	22.03
Netherlands	448	0.33	0.91	-26.80	-0.14	31.74
Norway	636	-0.91	1.13	-43.19	-1.11	36.78
Poland	1318	0.00	0.92	-32.82	0.03	30.04
Portugal	172	2.10	0.93	-29.66	0.47	27.47
Russia	529	-0.70	1.04	-31.64	1.06	42.44
South Africa	735	0.76	0.80	-17.66	0.66	19.34
South Korea	6359	0.30	0.92	-37.10	0.46	36.90
Spain	534	3.02	0.94	-31.75	1.78	33.97
Sweden	1433	-0.36	1.12	-45.30	-0.60	39.59
Switzerland	825	-0.78	0.95	-25.41	0.04	24.79
Turkey	1031	1.30	1.00	-36.34	0.57	34.75
USA	12535	-0.28	0.86	-31.32	0.02	31.42
United Kingdom	3479	0.57	0.93	-30.37	0.00	29.58
Panel B: Summary Statistics of Emission Growth across Sectors						
Agriculture, forestry, fishing	534	-0.10	1.05	-34.92	-1.04	35.47
Mining	3258	-0.60	1.14	-43.47	0.60	44.13
Construction	2569	-0.13	0.91	-28.52	0.51	30.43
Transportation & public utilities	6810	-1.12	0.96	-28.04	0.01	26.15
Wholesale trade	3577	0.35	1.18	-34.98	-0.19	37.58
Retail trade	4504	0.81	0.81	-26.93	-0.03	27.33
Services	11836	-0.62	0.89	-35.99	-0.50	34.04
Manufacturing sector	41641	0.02	**0.87**	-31.03	0.04	30.72
Full sample	74729	0.00	**0.91**	-31.46	0.03	31.33

Table A.1: Summary Statistics of Emission Growth (%)

A.3 Distance-to-Default

Figure A.4 plots the evolution of the average distance-to-default across countries over time. While the level of distance-to-default differs across countries, the series display substantial time variation within countries. In many cases, distance-to-default comoves across countries, suggesting the presence of common global financial factors, which we address by including year fixed effects.

Table A.2 reports summary statistics for distance-to-default across countries and sectors. Panel A documents substantial heterogeneity in average distance-to-default across countries, while Panel B shows systematic differences across industries. Importantly, the standard deviation within each country and sector is sizable, indicating meaningful variation in firms' credit conditions over time and across firms.

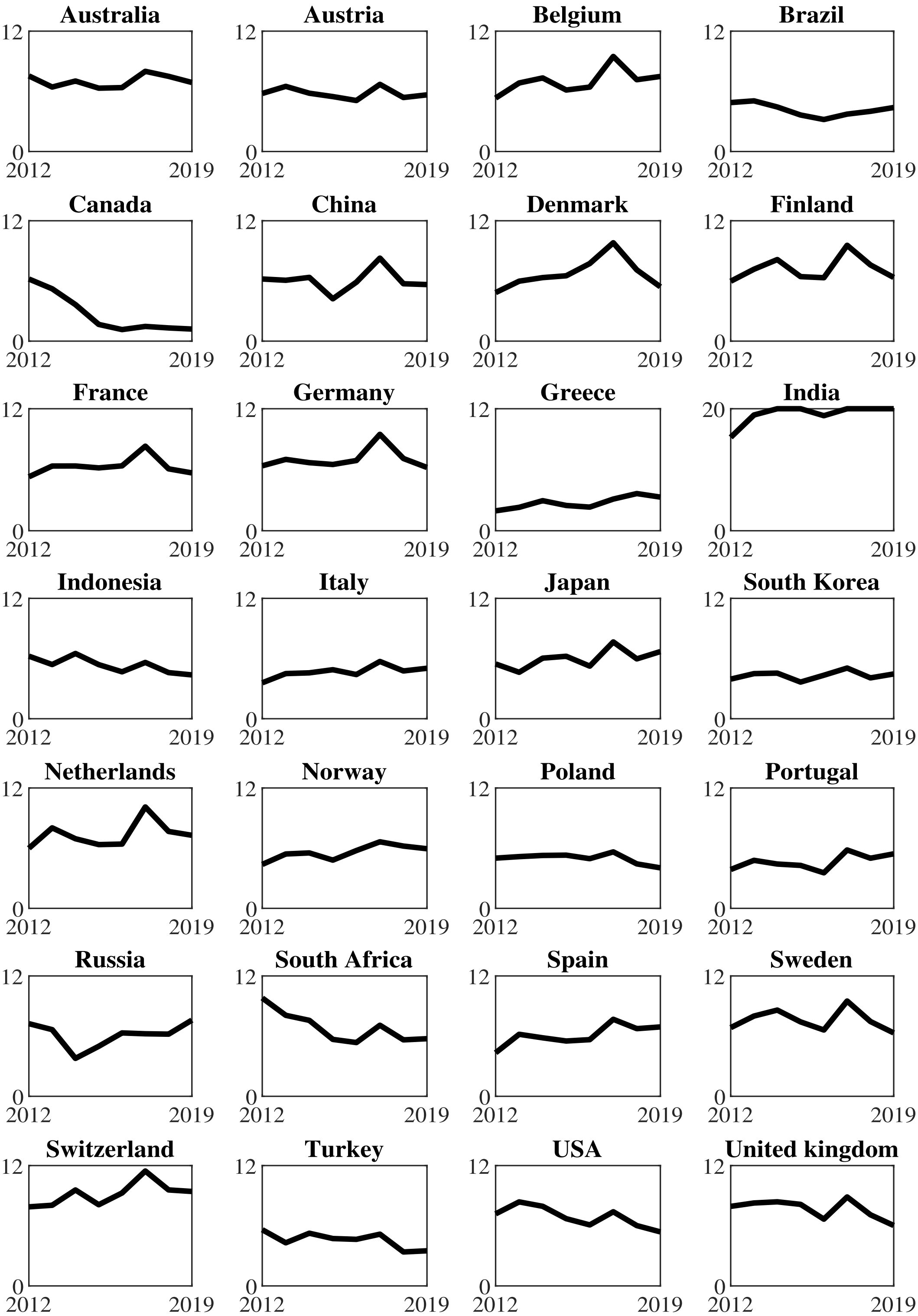


Figure A.4: Distance-to-Default over Time

	N	Mean	SD	P25	P50	P75
Panel A: Summary Statistics of D2D across Countries						
Australia	1584	8.53	5.8	4.32	6.89	11.01
Austria	268	7.29	4.42	4.57	5.93	8.20
Belgium	363	8.24	4.77	4.89	7.08	10.21
Brazil	747	5.24	4.13	2.58	4.12	6.76
Canada	1291	14.32	8.04	8.63	14.32	20.00
China	14420	7.95	5.61	4.01	5.85	9.77
Denmark	348	8.09	5.74	4.04	6.08	10.57
Finland	446	7.99	4.32	4.98	7.08	9.94
France	1742	7.53	4.45	4.60	6.34	9.07
Germany	1547	8.49	5.53	4.65	6.66	10.23
Greece	433	4.50	5.16	1.36	2.66	4.98
India	190	15.15	6.48	10.19	20.00	20.00
Indonesia	1724	8.75	6.93	3.19	5.47	15.74
Italy	646	5.74	3.45	3.48	4.82	7.15
Japan	14309	7.54	4.95	4.05	6.03	9.37
Netherlands	319	8.11	4.94	4.43	7.26	10.43
Norway	608	6.94	5.42	3.41	5.43	8.33
Poland	1064	6.21	4.60	3.29	4.92	7.46
Portugal	134	5.16	3.93	1.87	4.60	7.01
Russia	425	8.15	6.12	3.49	6.01	11.45
South Africa	688	8.29	5.12	4.57	7.05	10.99
South Korea	5364	6.04	4.95	3.01	4.32	6.77
Spain	474	7.50	5.16	3.99	5.92	9.68
Sweden	848	9.20	5.80	4.86	7.48	11.60
Switzerland	798	10.11	5.386	6.06	8.97	13.43
Turkey	1170	6.23	4.97	3.10	4.46	7.27
USA	8779	10.42	6.50	5.18	8.40	20.00
United Kingdom	2551	9.58	5.77	5.21	7.80	12.95
Panel B: Summary Statistics of D2D across Sectors						
Agriculture, forestry, fishing	454	8.11	5.40	4.26	6.14	10.20
Mining	1525	8.33	6.47	3.50	5.72	11.55
Construction	1793	6.70	4.87	3.33	5.26	8.58
Transportation & public utilities	4046	6.85	4.80	3.72	5.44	8.15
Wholesale trade	2492	7.56	5.35	3.88	5.70	9.36
Retail trade	2671	8.23	5.44	4.36	6.45	10.42
Services	5859	8.94	6.05	4.31	6.89	12.15
Manufacturing Sector	28538	7.84	**5.48**	3.93	5.94	9.88
Full Sample	47378	7.86	**5.50**	3.95	6.01	9.90

Table A.2: Summary Statistics of Distance-to-Default

A.4 Firm Controls

Summary statistics for the firm level control variables used in the empirical analysis are reported in Table A.3. Firm size (*Log(Assets)*) is measured as the natural logarithm of total assets. *young* is an indicator variable equal to one if the firm is younger than five years and zero otherwise. Profitability (*EBIT/Revenues*) is proxied by earnings before interest and taxes scaled by revenues. Capital intensity (*PPE/Assets*) is measured as the ratio of property, plant, and equipment to total assets. Liquidity (*CashFlow/Assets*) is proxied by cash flow scaled by total assets, while investment (*CapEx/Assets*) is measured by capital expenditures relative to total assets. The means and standard deviations of these variables are broadly comparable to those documented in the prior corporate finance literature (Almeida, Campello, and Weisbach, 2004), suggesting that our sample is representative along key firm characteristics. All variables are winsorized at the 1st and 99th percentiles.

Variables	N	Mean	SD	Min	P25	P50	P75	Max
Log(Assets)	79,068	9.00	2.93	2.38	7.06	8.94	11.02	16.02
Young	80,266	0.56	0.50	0.00	0.00	1.00	1.00	1.00
EBIT/Revenues	93,218	0.02	0.70	-5.79	0.05	0.10	0.18	0.63
PPE/Assets	93,267	0.61	1.08	0.00	0.12	0.27	0.58	7.28
CashFlow/Assets	92,408	0.14	0.14	0.00	0.04	0.10	0.19	0.71
CapEx/Assets	92,841	0.04	0.05	0.00	0.01	0.03	0.06	0.24
Leverage	84,495	0.24	0.24	0.00	0.04	0.17	0.38	1.00

Table A.3: Summary Statistics of Control Variables

B Additional Empirical Results

B.1 Exogeneity of Carbon Policy Shocks

A potential threat to our identification assumption is the endogeneity of carbon tax changes with respect to firm credit constraints. Policymakers might defer carbon tax hikes if the economy would otherwise experience financial distress (Döttling and Rola-Janicka, 2025). We test whether the probability of a carbon tax increase depends on *aggregate* credit constraints in country c in the previous year:

$$\text{Prob}(\text{Tax}_{c,t} \neq \text{Tax}_{c,t-1}) = \beta_0 + \beta_1 \cdot \text{D2D}_{c,t-1} + \beta_2 \cdot X_{c,t-1} + \epsilon_{c,t} \ . \tag{23}$$

Here $\text{D2D}_{c,t-1}$ refers to the aggregate distance-to-default in country c, where we use both median and average distance-to-default across firms in each country and year. The vector $X_{c,t-1}$ contains typical control variables including GDP growth, inflation rate, short-term and long-term interest rates, public debt-to-GDP ratio and unemployment rate.[12] We do not add country fixed effects, since we would loose observations in all countries without carbon tax changes. Standard errors are clustered at country level. The coefficient of interest is β_1: if it is different from zero, aggregate credit constraints would predict the probability of a tax change. We specify Equation (23) as a Probit-model, but find similar results in a Logit-model. The results in Table B.1 show that country-level aggregate credit constraints do not predict climate policy, irrespective of using the mean or median to aggregate firms within each country.

[12] We collect the public debt-to-GDP ratio from IMF Data Mapper. All the other control variables are obtained from OECD statistics.

VARIABLES	(1) Prob($\text{Tax}_{c,t} \neq \text{Tax}_{c,t-1}$)	(2) Prob($\text{Tax}_{c,t} \neq \text{Tax}_{c,t-1}$)
Mean D2D$(j, t-1)$	-0.015	
	(0.017)	
Median D2D$(j, t-1)$		-0.009
		(0.018)
Country-Controls	✓	✓
Observations	158	158
Pseudo R-squared	0.0544	0.0539

Table B.1: Credit Constraints and Carbon Tax Shocks at the Country Level. This table reports the results of estimating Equation (23). Column (1) refers to the mean of *D2D* for each country and column (2) to the median of *D2D*. Regressions are estimated at the country-year level. The regressions control for GDP growth, inflation rate, short-term and long-term interest rates, public debt-to-GDP ratio and unemployment rate, all lagged by one year. Standard errors, clustered at the country level, are in parentheses. ***, ** and * indicate significance at the 1%, 5% and 10% level, respectively.

B.2 Interacting Carbon Tax Shock with Control Variables

In this section, we present the results of adding interaction terms between firm level control variables and the carbon tax shock. If the tightness of credit constraints is correlated with firm size, age, and profitability, the key coefficient in our baseline specification might actually pick up heterogeneous responses of smaller, younger, or more profitable firms. To mitigate such concern, we estimate

$$\begin{aligned}\Delta \log(Emi)_{j,t} = \beta_0 &+ \beta_1 \cdot \mathrm{D2D}_{j,t-1} + \beta_2 \cdot \Delta\mathrm{Tax}_{c,t} + \beta_3 \cdot \mathrm{D2D}_{j,t-1} \times \Delta\mathrm{Tax}_{c,t} \\ &+ \beta_4 \cdot X_{j,t-1} + \beta_5 \cdot X_{j,t-1} \times \Delta\mathrm{Tax}_{c,t} + \chi_c + \tau_t + \epsilon_{j,t} \ . \end{aligned} \tag{24}$$

Table B.2 displays the results. Compared to the baseline results in Table 1, the coefficient on the interaction term $D2D_{j,t-1} \times \Delta Tax_{c(j),t}$ is even slightly larger and remains highly significant.

	(1) $\Delta log(Emi)_{j,t}$	(2) $\Delta log(Emi)_{j,t}$	(3) $\Delta log(Emi)_{j,t}$
$D2D_{j,t-1} \times \Delta Tax_{c(j),t}$	-0.002**	-0.006***	-0.006***
	(0.001)	(0.002)	(0.002)
$D2D_{j,t-1}$	0.005**	0.006**	0.006***
	(0.002)	(0.002)	(0.002)
$\Delta Tax_{c(j),t}$	-0.052**	-0.025	-0.022
	(0.025)	(0.036)	(0.028)
$Log(Assets)_{j,t-1}$	-0.022*	-0.028*	-0.034**
	(0.011)	(0.014)	(0.016)
$Young_{j,t-1}$	-0.021***	-0.024***	-0.019**
	(0.006)	(0.009)	(0.008)
$EBIT/Revenues_{j,t-1}$	-0.096***	-0.060***	-0.064***
	(0.020)	(0.020)	(0.019)
$Log(Assets)_{j,t-1} \times \Delta Tax_{c(j),t}$	0.003	0.008*	0.008**
	(0.002)	(0.005)	(0.004)
$Young_{j,t-1} \times \Delta Tax_{c(j),t}$	0.029	-0.000	-0.016
	(0.023)	(0.022)	(0.030)
$EBIT/Revenues_{j,t-1} \times \Delta Tax_{c(j),t}$	0.014***	0.019***	0.022***
	(0.003)	(0.006)	(0.008)
Constant	0.185*	0.237*	0.287**
	(0.096)	(0.120)	(0.140)
Observations	40,109	24,125	23,984
R-squared	0.024	0.020	0.111
Industry-by-year FE	2-digit SIC	NO	4-digit SIC
Country FE	YES	YES	YES
Sectors	All	Manuf	Manuf
Year FE	NO	YES	NO

Table B.2: Adding Interaction between Controls and Carbon Tax Shock. This table reports the results adding interactions between baseline controls and carbon tax shock as additional controls. Regressions are estimated at the firm-year level. The regressions control for firm size ($Log(Assets)_{j,t-1}$), age ($Young_{j,t-1}$), profitability ($EBIT/Revenues_{j,t-1}$) and their interactions with carbon tax shocks, all lagged by one year. We include country fixed effects in all specifications, year fixed effects in column (2) and industry × year fixed effects in column (1),where the industries are measured by 2-digit SIC, and column (3), where the industries are measured by the most granular 4-digit SIC. Standard errors, clustered at the country level, are in parentheses. ***, ** and * indicate significance at the 1%, 5% and 10% level, respectively.

B.3 Robustness: Alternative Controls

To further mitigate concerns that omitted firm characteristics may confound our baseline results, we augment the baseline specification with additional controls capturing firms' capital intensity, liquidity, and investment. Capital intensity is measured as the ratio of property, plant, and equipment (PP&E) to total assets, liquidity is proxied by the cashflow over total assets, and investment is measured by capital expenditures scaled by total assets. These variables account for heterogeneity in firms' production technology, financial flexibility, and real investment behavior that may be correlated with both $\Delta log(Emi)_{j,t}$ and $D2D_{j,t-1} \times \Delta Tax_{c(j),t}$. As reported in Table B.3, the coefficient of interest (β_1) is slightly larger in magnitude and remains statistical significance, indicating that our results are not driven by omitted variation in capital structure, liquidity conditions, or investment intensity.

VARIABLES	(1) $\Delta log(Emi)_{j,t}$	(2) $\Delta log(Emi)_{j,t}$	(3) $\Delta log(Emi)_{j,t}$	(4) $\Delta log(Emi)_{j,t}$	(5) $\Delta log(Emi)_{j,t}$	(6) $\Delta log(Emi)_{j,t}$
$D2D_{j,t-1} \times \Delta Tax_{c(j),t}$	-0.003**	-0.003**	-0.003**	-0.007***	-0.007***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
$D2D_{j,t-1}$	0.005**	0.007**	0.007**	0.007***	0.010***	0.010***
	(0.002)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)
$\Delta Tax_{c(j),t}$	-0.003	-0.003	-0.005	0.054***	0.053***	0.053***
	(0.008)	(0.007)	(0.007)	(0.013)	(0.013)	(0.014)
$Log(Assets)_{j,t-1}$	-0.012	-0.017	-0.019	-0.024	-0.027	-0.028*
	(0.012)	(0.013)	(0.013)	(0.016)	(0.016)	(0.016)
$Young_{j,t-1}$	0.007	0.013	0.009	0.010	0.016	0.010
	(0.008)	(0.008)	(0.007)	(0.009)	(0.009)	(0.008)
$EBIT/Revenues_{j,t-1}$	-0.090***	-0.098***	-0.124***	-0.060***	-0.077***	-0.092***
	(0.021)	(0.020)	(0.026)	(0.021)	(0.021)	(0.022)
$PPE/Assets$	0.011	0.001	-0.003	0.026*	0.014	0.011
	(0.009)	(0.008)	(0.008)	(0.014)	(0.011)	(0.008)
$CashFlow/Assets$		-0.385**	-0.386**		-0.471***	-0.485***
		(0.150)	(0.146)		(0.166)	(0.157)
$CapEx/Assets$			0.612***			0.492**
			(0.107)			(0.206)
Constant	0.102	0.190	0.187	0.183	0.261*	0.263*
	(0.094)	(0.115)	(0.116)	(0.126)	(0.149)	(0.146)
Observations	34,420	33,878	33,685	20,605	20,254	20,157
R-squared	0.020	0.021	0.023	0.107	0.109	0.111
Industry-by-year FE	2-digit SIC	2-digit SIC	2-digit SIC	4-digit SIC	4-digit SIC	4-digit SIC
Country FE	YES	YES	YES	YES	YES	YES
Sectors	All	All	All	Manuf	Manuf	Manuf

Table B.3: Alternative Controls. This table reports the results adding capital intensity, liquidity and investment as additional controls. Regressions are estimated at the firm-year level. The regressions control for firm size ($Log(Assets)_{j,t-1}$), age ($Young_{j,t-1}$), profitability ($EBIT/Revenues_{j,t-1}$), capital intensity ($PPE/Assets$), liquidity ($CashFlow/Assets$) or investment ($CapEx/Assets$), all lagged by one year. We include country fixed effects and industry $\times$ year fixed effects in all specifications. Column (1)-(3) record the results for all sectors and the industries are measured by 2-digit SIC. Column (4)-(6) for only manufacturing sector and the industries are measured by the most granular 4-digit SIC. Standard errors, clustered at the country level, are in parentheses. ***, ** and * indicate significance at the 1%, 5% and 10% level, respectively.

B.4 Robustness: Utilities

Throughout the analysis, we assume that companies respond to carbon tax increases in their country of incorporation. Since firms in our sample are large, it is reasonable to assume that they generate part of their emissions abroad. Such emissions would not respond directly to changes in domestic climate policy. To address this concern, Table B.4 shows that our baseline results also hold in Transportation and public utilities sector. These companies are usually not multinational and should mostly respond to domestic policies. The coefficient on the interaction term $D2D_{j,t-1} \times \Delta Tax_{c(j),t}$ remains negative and highly significant.

VARIABLES	(1) $\Delta log(Emi)_{j,t}$	(2) $\Delta log(Emi)_{j,t}$
$D2D_{j,t-1} \times \Delta Tax_{c(j),t}$	-0.004***	-0.003***
	(0.001)	(0.001)
$D2D_{j,t-1}$	0.000	-0.000
	(0.001)	(0.001)
$\Delta Tax_{c(j),t}$	-0.004	-0.023
	(0.022)	(0.023)
Controls	✓	✓
Observations	3,237	3,233
R-squared	0.033	0.076
Industry-by-year FE	NO	YES
Country FE	YES	YES
Year FE	YES	NO
Sample	Transportation & Public Utilities	Transportation & Public Utilities

Table B.4: Carbon Taxes and Credit Constraints: Utility Sector. This table reports the baseline results in the sector of Transportation and Public Utilities. Regressions are estimated at the firm-year level. The regressions control for firm size ($Log(Assets)_{j,t-1}$), age ($Young_{j,t-1}$), profitability ($EBIT/Revenues_{j,t-1}$), all lagged by one year. We include country fixed effects in all specifications, year fixed effects in column (1) and industry × year fixed effects in column (2),where the industries are measured by 4-digit SICs. Standard errors, clustered at the country level, are in parentheses. ***, ** and * indicate significance at the 1%, 5% and 10% level, respectively.

B.5 Emission Growth and Carbon Tax by Credit Constraints

In a simple specification (Equation (25)), we test whether firms' emission growth responds differently to changes in carbon taxes depending on their credit constraints, proxied by firms' distance-to-default and classified as high or low relative to the sectoral median. We regress emission growth on changes in country-specific carbon taxes, estimating Equation (25) separately for firms with high and low credit constraints.

$$\Delta \log(Emi)_{j,t} = \beta_0 + \beta_1 \Delta \text{Tax}_{c(j),t} + \beta_2 \cdot X_{j,t-1} + \chi_c + \tau_t + \epsilon_{j,t} \ . \tag{25}$$

Table B.5 shows that emission growth responds more strongly to carbon-tax shocks for firms with higher distance-to-default, i.e., less financially constrained firms. This pattern is robust to controlling for capital intensity, although its magnitude is attenuated once capital intensity is included. Quantitatively, the difference is comparable to the specification with an interaction term between distance-to-default and the carbon tax change.

VARIABLES	(1) $\Delta log(Emi)_{j,t}$	(2) $\Delta log(Emi)_{j,t}$	(3) $\Delta log(Emi)_{j,t}$	(4) $\Delta log(Emi)_{j,t}$
$\Delta\, Tax_{c(j),t}$	-0.047***	-0.037***	-0.038***	-0.033***
	(0.011)	(0.009)	(0.012)	(0.010)
$Log(Assets)_{j,t-1}$	-0.018*	-0.053**	-0.018	-0.051**
	(0.011)	(0.022)	(0.011)	(0.021)
$Young_{j,t-1}$	-0.004	-0.017	0.002	-0.022
	(0.013)	(0.015)	(0.014)	(0.014)
$EBIT/Revenues_{j,t-1}$	-0.073	-0.111***	-0.068*	-0.129***
	(0.043)	(0.032)	(0.039)	(0.033)
$CapIntensity_{j,t-1}$			-0.029**	0.018*
			(0.013)	(0.010)
Constant	0.152	0.551**	0.165	0.532**
	(0.092)	(0.219)	(0.101)	(0.206)
Observations	19,541	19,506	16,690	16,632
R-squared	0.175	0.176	0.187	0.191
Industry-by-year FE	2-digit SIC	2-digit SIC	2-digit SIC	2-digit SIC
Country FE	YES	YES	YES	YES
Sectors	Full	Full	Full	Full
D2D	High	Low	High	Low

Table B.5: Emission Growth and Carbon Taxes for Constrained and Unconstrained Firms. This table reports the results of estimating Equation (25). Columns (1) and (2) refer to the specification in sub-sample firms with a high and low distance-to-default, obtained from a median split within each industry, without controlling for capital intensity. Columns (3) and (4) provide the results with capital intensity as an additional control. $\Delta\, Tax_{c,t}$ is the difference in country-level taxes from $t-1$ to t. The regressions control for firm size ($Log(Assets)_{j,t-1}$), age ($Young_{j,t-1}$) and profitability ($EBIT/Revenues_{j,t-1}$), all lagged by one year. We include country fixed effects and industry $\times$ year fixed effects in all specifications, where the industries are defined as their 2-digit SIC codes. Standard errors, clustered at the country level, are in parentheses. ***, ** and * indicate significance at the 1%, 5% and 10% level, respectively

C Model Appendix

This section presents additional details on the maximization problem of manufacturing good firms.

Abatement Effort The first-order condition w.r.t a_t reads

$$\big(1-G(\overline{m}_t)\big)\tau_t k_t - G'(\overline{m}_t)\frac{\partial \overline{m}_t}{\partial a_t}\big(p_t^Z - \tau_t(1-a_t)\big)k_t + F'(\overline{m}_t)\frac{\partial \overline{m}_t}{\partial a_t}\chi l_t = \alpha_0 a_t^{\alpha_1} k_t \ . \tag{C.1}$$

The first term of Equation (C.1) reflects the expected savings on carbon taxes ($\tau_t k_t$), conditional on not defaulting $(1-G(\overline{m}_t))$. The second term captures the marginal increase in (after-tax revenues) associated with a lower default threshold. By contrast, the third part takes into account that a lower default threshold also increases the expected debt repayment obligation. The partial derivative of the default threshold, Equation (5), with respect to a_t is given by

$$\frac{\partial \overline{m}_t}{\partial a_t} = -\frac{\chi l_t}{(p_t^Z-(1-a_t)\tau_t)^2 k_t}\tau_t = -\frac{\tau_t \overline{m}_t}{\widetilde{p}_t^Z} \tag{C.2}$$

such that we can write the first-order condition as

$$\begin{aligned}
&\big(1-G(\overline{m}_t)\big)\tau_t k_t + G'(\overline{m}_t)\frac{\tau_t\overline{m}_t}{\widetilde{p}_t^Z}\widetilde{p}_t^Z k_t - F'(\overline{m}_t)\frac{\tau_t\overline{m}_t}{\widetilde{p}_t^Z}\chi l_t \frac{k_t}{k_t} = \alpha_0 a_t^{\alpha_1} k_t \ , \\
&\big(1-G(\overline{m}_t)\big)\tau_t k_t + G'(\overline{m}_t)\tau_t\overline{m}_t k_t - F'(\overline{m}_t)\tau_t\overline{m}_t^2 k_t = \alpha_0 a_t^{\alpha_1} k_t \ , \\
&\big(1-G(\overline{m}_t)\big)\tau_t = \alpha_0 a_t^{\alpha_1} \ .
\end{aligned}$$

where we have used that $G'(\overline{m}_t) = F'(\overline{m}_t)\overline{m}_t$. Intuitively, there is a positive and a negative marginal effect of abatement on the default threshold that exactly offset each other because the default threshold represents the productivity draw at which firms can *exactly* repay their creditors. Re-arranging yields Equation (9).

Investment, Debt and Risk Choice The Lagrangian associated with the inter-period maximization problem reads

$$\begin{aligned}\mathcal{L}_t = & -p_t^K k_{t+1} - \frac{\alpha_0}{1+\alpha_1}\left(a_{t+1}\right)^{1+\alpha_1} + q(\overline{m}_{t+1})\Big(l_{t+1} - (1-\chi)l_t\Big) \\ & + \mathbb{E}_t\Bigg[\widetilde{\Lambda}_{t,t+1}\cdot\Bigg\{\int_{\overline{m}_{t+1}}^{\infty}\big(p_{t+1}^Z - \tau_{t+1}(1-a_{t+1})\big)m_{t+1}k_{t+1} - \chi\cdot l_{t+1}dF(m_{t+1}) \\ & \quad + p_{t+1}^K(1-\delta_K)k_{t+1} + q(\overline{m}_{t+2})\Big(l_{t+2} - (1-\chi)l_{t+1}\Big)\Bigg\}\Bigg] + \\ & + \mu_t\left(\overline{m}_{t+1} - \frac{\chi l_{t+1}}{(p_{t+1}^Z - \tau_{t+1}(1-a_{t+1}))k_{t+1}}\right) .\end{aligned}$$

Differentiating with respect to k_{t+1}, l_{t+1} and $\overline{m}_{t+1}$ and the multiplier λ_t yields equations (11), (12) and (13). As pointed out by Gomes, Jermann, and Schmid (2016), this period's risk choice depends on next period's risk choice $\overline{m}_{t+2}$ in the presence of long-term debt. Following Giovanardi and Kaldorf (2025), we pin down the slope of the policy function for $\overline{m}_{t+1}$ by further differentiating a first-order condition associated with the shareholder value maximization problem. We re-arrange the first-order condition for loans, Equation (12), for the Lagrangian multiplier λ_t on the risk choice,

$$\lambda_t = \frac{l_{t+1}}{\overline{m}_{t+1}}q(\overline{m}_{t+1}) - \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\frac{l_{t+1}}{\overline{m}_{t+1}}\Big(\chi(1-F(\overline{m}_{t+1})) + (1-\chi)q(\overline{m}_{t+2})\Big) , \tag{C.3}$$

and plug this expression into the first-order condition for capital (11) to get:

$$\begin{aligned} p_t^K - & \mathbb{E}_t\frac{l_{t+1}}{k_{t+1}}\Big(q(\overline{m}_{t+1}) - \widetilde{\Lambda}_{t,t+1}\left\{\chi(1-F(\overline{m}_{t+1})) + (1-\chi)q(\overline{m}_{t+2})\right\}\Big) \\ = & \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big((1-\delta_k)p_t^K + (1-G(\overline{m}_{t+1}))\widetilde{p}_{t+1}^Z\Big) .\end{aligned}$$

Multiplying both sides by $\frac{\chi}{\widetilde{p}_{t+1}^Z}$,

$$\begin{aligned} \frac{\chi}{\widetilde{p}_{t+1}^Z}p_t^K - & \overline{m}_{t+1}\mathbb{E}_t\Big(q(\overline{m}_{t+1}) - \widetilde{\Lambda}_{t,t+1}\left\{\chi(1-F(\overline{m}_{t+1})) + (1-\chi)q(\overline{m}_{t+2})\right\}\Big) \\ = & \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\left(\frac{\chi}{\widetilde{p}_{t+1}^Z}(1-\delta_k)p_{t+1}^K + \chi(1-G(\overline{m}_{t+1}))\right) ,\end{aligned}$$

and differentiating this expression w.r.t. $\overline{m}_{t+1}$, we obtain

$$\begin{aligned} & -q(\overline{m}_{t+1}) + \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\left(\chi(1-F(\overline{m}_{t+1})) + (1-\chi)q(\overline{m}_{t+2})\right) \\ & -\overline{m}_{t+1}\mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\left(\frac{\partial q_t}{\partial\overline{m}_{t+1}} + \chi f(\overline{m}_{t+1}) + (1-\chi)\frac{\partial q_{t+1}}{\partial\overline{m}_{t+2}}\frac{\partial\overline{m}_{t+2}}{\partial\overline{m}_{t+1}}\right) = -\chi\mathbb{E}_t\widetilde{\Lambda}_{t,t+1}G'(\overline{m}_{t+1}) .\end{aligned}$$

We can simplify this expression by exploiting that the derivative of the conditional mean $G'(\overline{m}_{t+1})$ satisfies $G'(\overline{m}_{t+1}) = \overline{m}_{t+1}F'(\overline{m}_{t+1})$:

$$q(\overline{m}_{t+1}) + \overline{m}_{t+1}\frac{\partial q_t}{\partial \overline{m}_{t+1}} = \mathbb{E}_t\widetilde{\Lambda}_{t,t+1}\Big(\chi(1-F(\overline{m}_{t+1})) + (1-\chi)\Big\{q(\overline{m}_{t+2}) + \overline{m}_{t+1}\frac{\partial q_{t+1}}{\partial \overline{m}_{t+2}}\frac{\partial \overline{m}_{t+2}}{\partial \overline{m}_{t+1}}\Big\}\Big)\ , \tag{C.4}$$

is an additional equilibrium condition that pins down the slope of the policy function for leverage $\frac{\partial \overline{m}^f_{t+2}}{\partial \overline{m}_{t+1}}$. Intuitively, the LHS summarizes the additional dividends today financed by issuing one unit of loans net of a debt dilution term, as $q'(\overline{m}_{t+1})$ is negative. The RHS indicates that increasing leverage today increases the expected repayment obligation from the maturing loan share χ and also affects how a higher leverage affects the additional dividends tomorrow that can be financed by issuing one unit of loans.

Additional Equilibrium Conditions Labor and intermediate good demand by competitive final good firms is given by:

$$w_t = (1-\theta)\frac{y_t}{n_t} \quad \text{and} \quad p^Z_t = \theta\frac{y_t}{z_t}\ , \tag{C.5}$$

$$w^{rf}_t = (1-\theta)\frac{y^{rf}_t}{n^{rf}_t} \quad \text{and} \quad p^{Z,rf}_t = \theta\frac{y^{rf}_t}{z^{rf}_t}\ . \tag{C.6}$$

Separating the labor demand by constrained and unconstrained firms from their investment and leverage choices allows for a more straightforward exposition of financial frictions and makes the aggregation step more straightforward, but it is not a crucial assumption. The household first-order conditions for labor and deposits read:

$$\frac{w_t}{c_t} = \omega_N n_t\ , \tag{C.7}$$

$$\frac{w^{rf}_t}{c_t} = \omega_N\big(n^{rf}_t\big)\ , \tag{C.8}$$

$$1 = \Lambda_{t,t+1}(1+r_t)\ , \tag{C.9}$$

where $\Lambda_{t,t+1} \equiv \beta\frac{c^{-1}_{t+1}}{c^{-1}_t}$ denotes the representative household's stochastic discount factor.

Response to carbon tax Shock with Persistent Abatement In Figure C.1, we display the differential effects of a 10\$/tCO2 carbon tax shock on abated emissions and the emission reduction in the extension with persistent abatement. The depreciation rate of abatement investment is set to $\delta_A = 0.08$, so that the share of abated emissions only increases gradually, which is shown in the upper right panel of Figure C.1. Consequently, emissions decline much more slowly, and the discrepancy between safe/unconstrained and

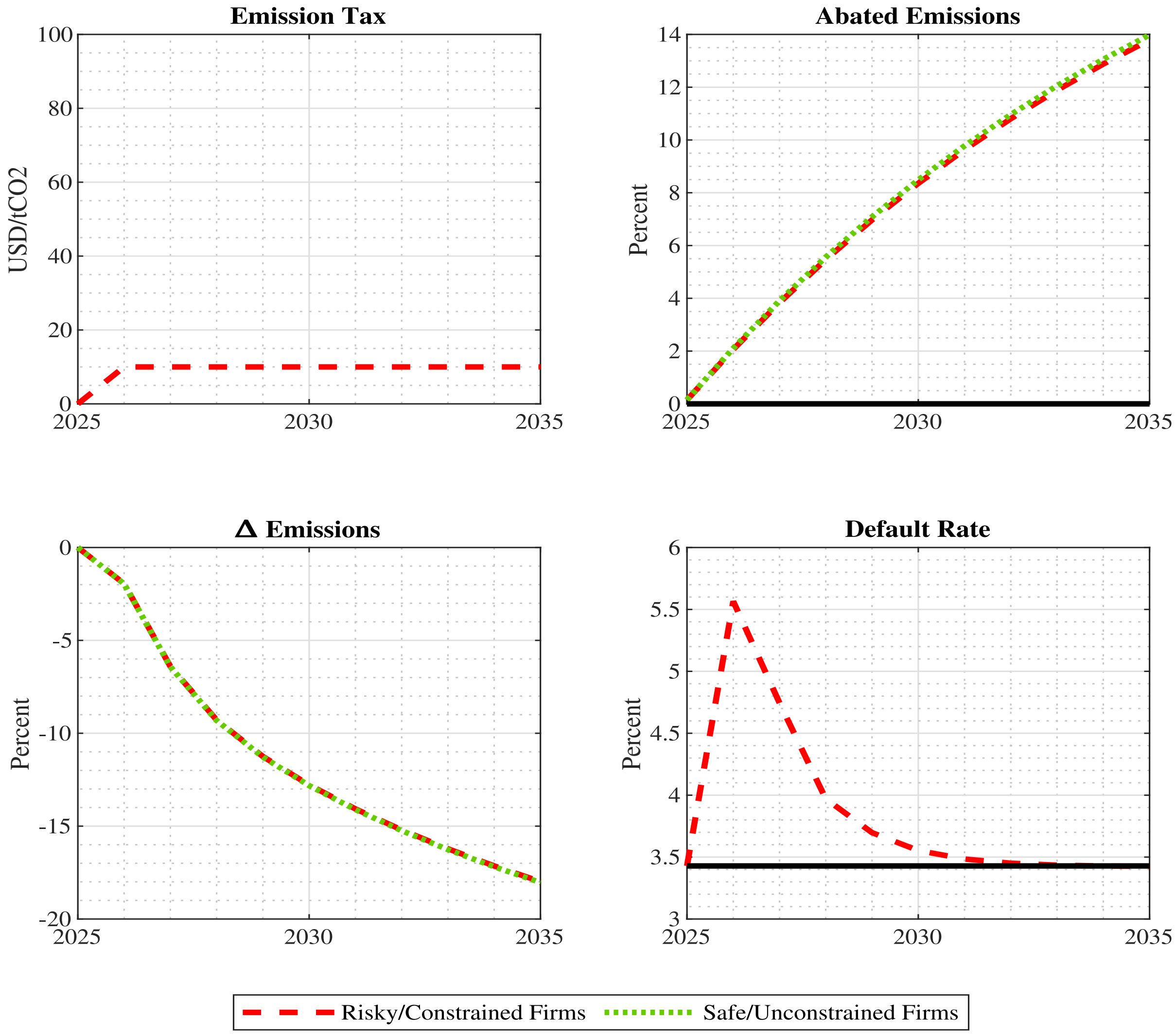


Figure C.1: Transmission of Carbon Taxes with Persistent Abatement. This figure displays the dynamic effects of an unanticipated and permanent carbon tax shock of 10\$/tCO2 (top left). The share of abated emissions for risky/constrained (dashed red line) and safe/unconstrained (dotted green line) firms is indicated in the top right. The bottom left shows the emission reductions for both firm types. The bottom right shows the default rate for risky/constrained firms.

risky/constrained is only visible with a substantial lag. Lastly, the default rate of risky firms responds less strongly to the carbon tax shock than in the baseline economy.